Autonomous Coastal Profiler
Project Final Report

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Final Year Project Team M16

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1. Executive Summary

The Cawthron Institute wants to monitor oceanic water parameters, but existing commercial options are either too expensive or have inadequate performance. They approached the University of Canterbury and commissioned a Final Year Project to develop a low-cost reliable ocean profiling method for deployment in Tasman Bay. The system is a motorised ‘profiler’ which travels along a vertical mooring cable suspended beneath a buoy. The profiler has sensors for capturing data, which is transmitted to the buoy when it docks for recharging. The buoy is fitted with solar panels for recharging the profiler, and a cellular modem for transmitting data to shore.

The project team has produced a working prototype, which will need further testing to characterise its performance before being commercially viable. This prototype has passed underwater motion tests in a swimming pool and is watertight. A proof-of-concept system for Bluetooth control and communications has been developed, but this will need further development with hardware-in-the-loop before autonomous deployment. A niobium connector has also been developed as an alternative to the wireless systems, but this will need underwater testing. The project spent $2799.34, with Cawthron provided components costing an additional $2669.00 for a final cost of $5468.34. The next steps should be to validate the profiler’s long term deployment requirements. Ultimately, the project has produced a profiler prototype which is well on the way to meeting all the design requirements and functioning as desired by Cawthron, for a fraction of the price of existing commercial solutions.

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3. Introduction

In the face of a changing climate, it is increasingly important to develop an understanding of how ecosystems respond to environmental changes. Oceans are affected by pollution, acidification, and temperature increases which alter delicate ecosystems (Hitz & Smith, 2004). Thus, local governments and the aquaculture industry are increasingly interested in understanding the waters surrounding New Zealand. The Cawthron Institute provides bespoke water-monitoring solution for parameters such as temperature, photosynthetic active radiation (PAR), salinity, and dissolved oxygen levels.

Cawthron aims expand their services with a new underwater monitoring system, but existing commercial options do not meet their requirements (M16 Project Team, 2024a). Consequently, they have commissioned this University of Canterbury Final Year Project to investigate, design, build, and test an underwater profiling system. The project aims to deliver a low-cost robot for water column profiling. If successful, it may be deployed Tasman Bay over a 12-month period.

Table 1 contains the requirements for this project which includes constraints and specifications. This end-of-year report details the achievements to date, and a brief discussion of design history and methodology. An updated timeline and budget are presented, and recommendations for both the client and any future developers are discussed.

Table 1: The requirements the final product must meet to be successful.

No.	Section	Requirement	Priority
1	Functional	The device shall measure depth and temperature of a water column up to 30 metres deep for near shore profiles.	Hard
2	Functional	The device should be able to measure water salinity and PAR.	Soft
3	Functional	The device shall take measurements of each parameter at intervals no less than every half a metre deep.	Hard
4	Functional	The device shall use no more than two sensors for each measured parameter.	Hard
5	Functional	The device shall control its position to within $\pm$ 200 mm.	Hard
6	Functional	The device shall be able to sample each desired measurement point at least once per hour for its entire deployment duration.	Hard
7	Functional	The device shall take fewer than 20 minutes to sample the water column parameters as described in Requirement 6.	Hard
8	Functional	The device shall be rechargeable from buoy-mounted batteries.	Hard
9	Functional	The device shall transmit its measurement data to the recharging buoy.	Hard
11	Functional	The device shall be installable by a single diver with a maximum of one hand tool.	Hard
12	Functional	The device shall operate autonomously for no less than six months.	Hard
13	Non-Functional	The device shall be deployable from Cawthron’s boat.	Hard
14	Non-Functional	The device shall support the addition of further sensors.	Hard
15	Non-Functional	The device shall be positively buoyant.	Hard

No.	Section	Requirement	Priority
16	Non-Functional	The device should have an emergency recovery mechanism which will propel it to the surface.	Soft
17	Non-Functional	The device shall weigh less than 15 kg to enable easy transportation.	Hard
18	Quality	The device shall be able to run off its own battery power for 72 hours of standard operation as outlined in Requirement 6.	Hard
19	Constraint	The prototype shall have component costs of fewer than $4000, excluding sensors.	Hard

4. Scope Changes

There have been no scope changes since the mid-year report (M16 Project Team, 2024b).

5. Achievements

This achievements section has been divided by subsystem, each having their own success criteria from the requirements in Table 1. The system overview section will discuss the global achievements and requirements satisfied by the final design. The other sections will focus on the subsections which contributed to the final design.

5.1. System Overview

Figure 1 shows a side view of the final prototype design, along with the profiler midway through an underwater test.

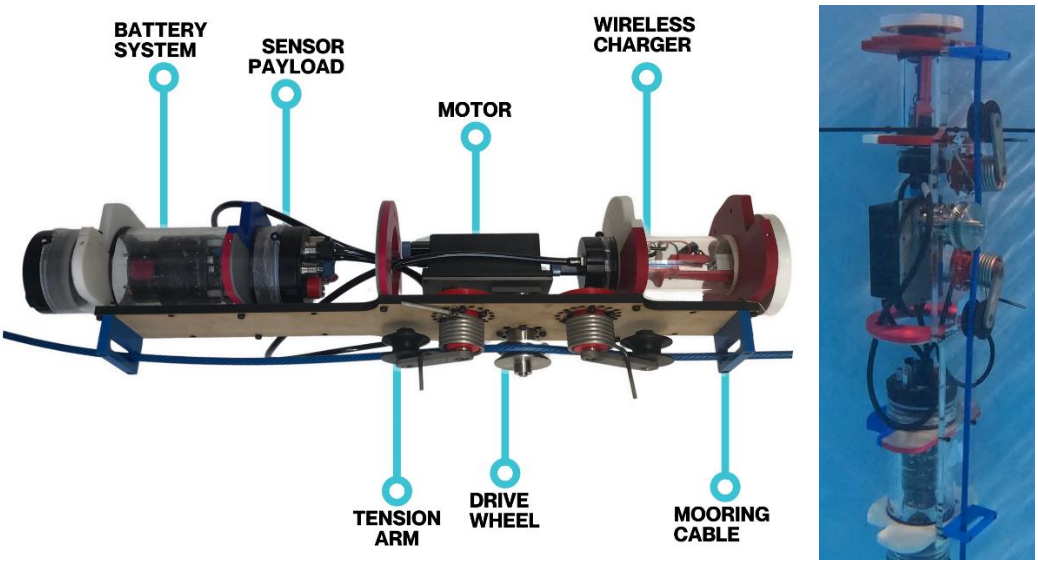

Figure 1: A side view of the most recent profiler model (left) and the prototype underwater (right).

5.2. Major Milestones and Test Results

Throughout the project major project milestones have been completed by integrating multiple systems to complete a task. These milestones either verified design requirements were met or made significant progress towards meeting them. These tests are used to justify the design's completion of the requirements outlined in Table 1.

Crawling tests were the first major milestone, where the tensioner arms, drive wheel, motor, and battery system worked together to drive the robot along a cable. Figure 2 shows an early prototype of the tensioner mechanism and drive wheel using battery power. The structure successfully crawled back and forth horizontally between two tables. This initial test confirmed each system was appropriate for the task and could progress to manufacturing.

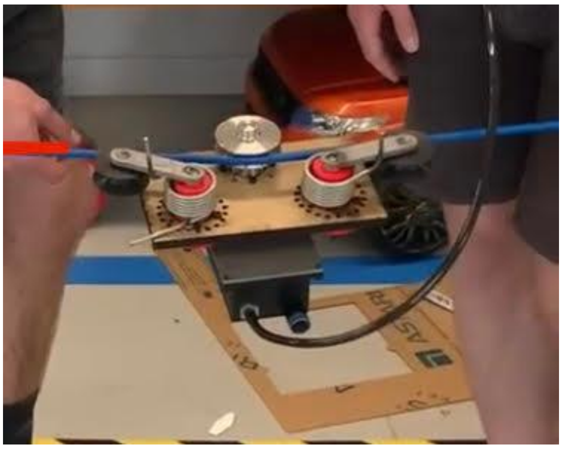

Figure 2: Horizontal test of a prototype crawling mechanism and motor housing.

Figure 3 shows a second test with a prototype chassis and every major component. This vertical climb test was conducted in the mechanical workshop. The profiler was counterbalanced with a pulley system to simulate neutral buoyancy. It successfully crawled up to three metres off the ground before descending again. This test verified robot was able to climb without the chassis flexing, which was a concern due to the potential for fatigue to reduce the lifespan of the profiler. After this test a second chassis was made from acetal. This test verified that Requirements 3, 5, and 7 were either satisfied or had a high probability of being satisfied (Table 2).

Figure 3: Profiler completing a vertical test with a protype chassis, which was then made from acetal.

Each pressure vessel had its depth rated via vacuum tests. Vessels succeeded if they could sustain a pressure differential of 1 bar of pressure, which is equivalent to being submerged under 10 metres of water. Each housing was connected to a vacuum pump, which pulled a vacuum inside the housings (Figure 4). The interior pressure was measured after 5 minutes. If the vessel had not repressurised during this time, it was considered safe for underwater testing with a low risk of failure.

Figure 4: Vacuum testing of each of the housings.

After every vessel passed the vacuum test, the combined system was deployed underwater at Jellie Park Pool in Christchurch. The test was limited to a depth of two metres by the available facilities. The robot was able to move without leaking while controlling its speed and position underwater. This test provided strong evidence for the satisfaction of Requirements 1, 3, 5, 6, 7, 11, and 13. **Video footage of this test can be found here:** https://youtu.be/KRtfzJgC5ew

5.3. Requirement Verification

Almost all requirements have been tested to validate the design. A list of the minor tests performed is included below and more details are available in the relevant design section.

- Battery recharges at adequate speed to sample every hour (Section 5.6).
- 2.4 GHz range in saltwater is suitable for underwater communications (Section 5.7).
- Out of water, vertical, orientation mechanism test using weights to simulate neutral buoyancy (Section 5.10).
- Temperature sensor responds to minor temperature fluctuations from body heat (Section 5.11).

Table 2 shows requirements and their degree of satisfaction. Unfortunately, most requirements need a fully autonomous deployment over several days to be completely satisfied. This has not yet been completed due to resource and time constraints. This means there is confidence that the design will meet most of the requirements, but no certainty.

Table 2: The global requirements which have been tested and met.

No.	Section	Requirement	Priority	Met?
1	Functional	The device shall measure depth and temperature of a water column up to 30 metres deep for near shore profiles.	Hard	Not fully tested. Conducted tests indicate passing.
2	Functional	The device should be able to measure water salinity and PAR.	Soft	No. This requirement was deemed unnecessary to proof-of-concept success by Cawthron.
3	Functional	The device shall be able to take measurements of each parameter at intervals no less than every half a metre deep in the water column.	Hard	Not fully tested. Conducted tests indicate passing.
4	Functional	The device shall use no more than two sensors for each measured parameter.	Hard	Yes
5	Functional	The device shall control its position to within $\pm$ 200 mm.	Hard	Yes
6	Functional	The device shall be able to sample each desired measurement point at least once per hour for its entire deployment duration.	Hard	Not fully tested. Conducted tests indicate passing.
7	Functional	The device shall take fewer than 20 minutes to sample the water column parameters as described in Requirement 6.	Hard	Not fully tested. Conducted tests indicate passing.
8	Functional	The device shall be rechargeable from buoy-mounted batteries.	Hard	Not fully tested. Conducted tests indicate passing.
9	Functional	The device shall transmit its measurement data to the recharging buoy.	Hard	Not fully tested. Conducted tests indicate passing.
11	Functional	The device shall be installable by a single diver with a maximum of one hand tool.	Hard	Yes
12	Functional	The device shall operate autonomously for no fewer than six months.	Hard	Untested
13	Non-Functional	The device shall be deployable from Cawthron's boat.	Hard	Not fully tested. Conducted tests indicate passing.
14	Non-Functional	The device shall support the addition of further sensors.	Hard	Yes
15	Non-Functional	The device shall be positively buoyant.	Hard	Yes
16	Non-Functional	The device should have an emergency recovery mechanism which will propel it to the surface.	Soft	No. This requirement was deemed unnecessary to proof-of-concept success by Cawthron.
17	Non-Functional	The device shall weigh less than 15 kg to enable easy transportation.	Hard	Yes
18	Quality	The device shall be able to run off its own battery power for 72 hours of standard operation as outlined in Requirement 6.	Hard	Untested
19	Constraint	The prototype shall have component costs of fewer than $4000, excluding sensors.	Hard	Yes

5.4. Chassis Design Achievements

5.4.1. Component Layout

The chassis supports the profiler's functional components. It incorporates the tensioning mechanism, drive train, charging, communication, and data processing systems, as illustrated in Figure 5. The charging and communication systems are positioned at the top of the chassis, as proximity to the buoy is essential for effective charging and data transfer. The drive train is situated in the middle of the chassis to evenly space the idler wheels around the drive wheel. Mooring line guides are placed at the top and bottom of the chassis to limit movement of the profiler perpendicular to the mooring line. Additionally, the chassis will use floats to obtain neutral buoyancy, minimising the energy to move through the water. These floats will be made from buoyant foam strategically fitted around components to minimize the system size.

The profiler has a longitudinal arrangement to fit through a 260 mm diameter opening in the centre of Cawthron's buoys for installation. The chassis has been constructed from acetal rather than the initially proposed stainless steel to reduce the amount of buoyant foam needed. It features a rib-like structure with a main plate and perpendicular plates for clamping components and supporting a hydrodynamic shell, as shown in Figure 5 and 6. The shell is cylindrical with an outer diameter of 190 mm, and the total length of the chassis is 640 mm.

Load testing revealed the acetal main plate is sufficiently strong and has minimal deflection, even in the absence of additional support bars. The chassis includes additional holes around the tensioner mechanism for spring tension adjustments, and spare holes for mounting future components.

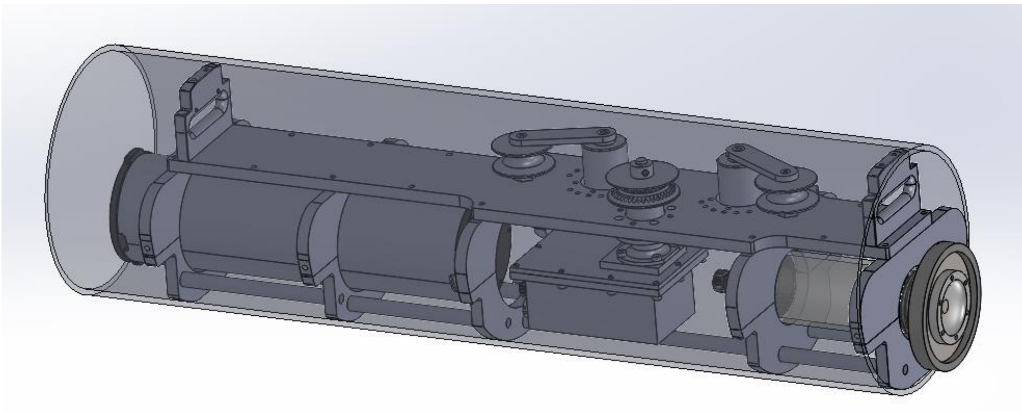

Figure 5. Full profiler assembly.

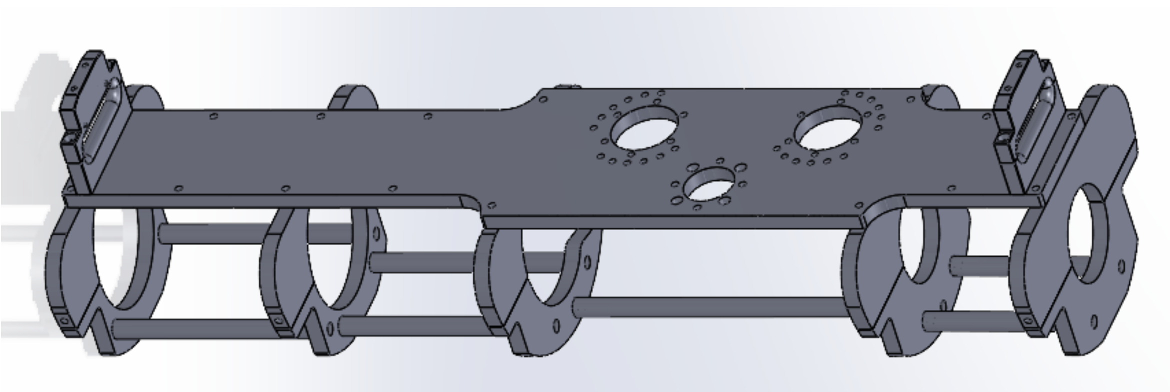

Figure 6. Chassis layout, with optional supporting bars.

5.4.2. Mooring Line Guides

The mooring line guides tether the profiler to the mooring line. They direct the mooring line and constrain the profiler's movement perpendicular to the tension mechanism. Initial concepts in Appendix A used a hoop design; however, these designs failed to consider movement within the tensioner system. The final design is machined from acetal and depicted in Figure 7. Acetal was chosen for its lightweight and cost-effectiveness compared to stainless steel (Samyn & Patrick De Baets, 2005).

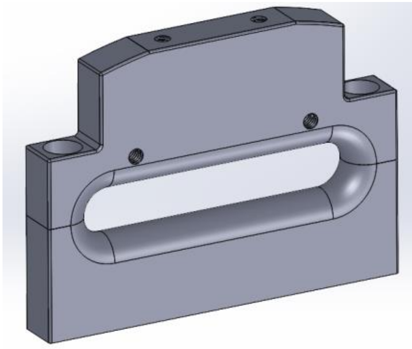

Figure 7. The CAD model of mooring line guides.

This guide performs satisfactorily, letting the mooring line move with the articulating tensioner while keeping the profiler aligned with the cable. Friction between the guides and cables is minimal because of acetal's low surface roughness. However, attaching the guide to the line can be challenging, as the same bolts which secure the two halves together also connect the guide to the chassis. Alternative options for clipping systems could be considered for future improvements.

5.5. Crawling Mechanism Achievements

5.5.1. Wheels and Pulleys

Initial concepts for the crawling mechanism explored various configurations of wheels, pulleys, and a claw system, as detailed in Appendix B. Cawthron advised against using the claw mechanism, as it could damage the cable's plastic jacketing. Additionally, Cawthron required a mechanism capable of changing direction. Profilers that implement claw mechanisms which are capable of changing direction are mechanically complicated, so a pulley system was selected for further development. This system consists of three wheels, as shown in Figure 8. The left and right wheels function as tension idlers, guiding the mooring line and maintaining adequate tension on the drive wheel for optimal grip. The middle wheel serves as the drive wheel, propelling the profiler along the mooring line.

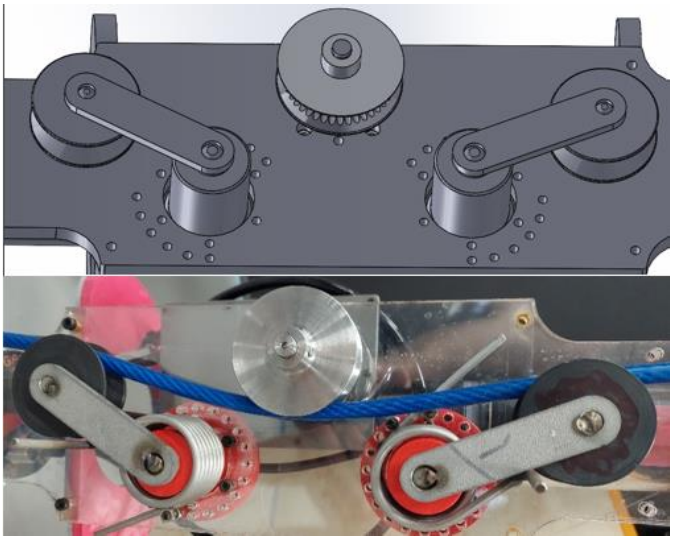

Figure 8. The final tensioner mechanism.

Research on the pulley groove design revealed that a V-groove with an internal angle of 70 degrees would yield the highest friction coefficient (Hrabovský, Kudrna, & Gaszek, 2022). The wheel uses radial slots to maximise friction. The guide wheels are made from acetal to reduce weight. The drive wheel is made from stainless steel to support higher loads and reduce wear on its corners.

The guide wheels operate on internal bearings, which must be sealed from particulate matter in the saltwater environment to extend their lifespan. Ceramic bearings were considered for this application because they are lubrication-free and corrosion-resistant ("SMB Bearings," 2024). However, due to long lead times they were replaced with acetal bearings featuring 316 SS balls, which offer similar advantages. Lubrication is still necessary for the seals' longevity so the wheel housing must contain lubricant and prevent contaminant ingress. To accomplish this, one side of the housing is sealed with a threaded fastener, while the other features a radial seal. The threaded fastener is sealed using an O-ring face seal with the bearing shaft being sealed with an X-profile O-ring rated to 3 bar of pressure. These were selected for the radial seals because they were cheaper and smaller than traditional O-rings. The two leftmost images in Figure 9 display the CAD model of the guide wheel, while the right image shows the final product made from acetal. Figure 10 shows the acetal bearing along with the 316 SS pin that the bearing runs on. 316 SS was used because it is corrosion resistant and is welded to the tensioner arm. This enabled the design and development of the final tensioner arms, which are depicted in Figure 8.

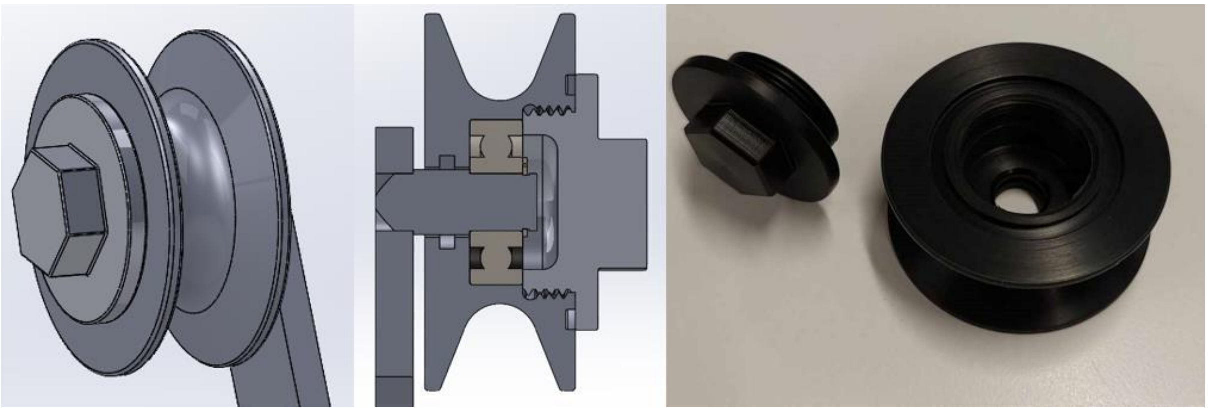

Figure 9: The final guide wheel design.

Figure 10. Guide wheel pin and bearing.

The guide wheel assembly operates as intended. There is minimal friction between the mooring line and the guide wheels. However, the wheels have excessive wobble because the bearing raceways are not adequately constrained by the stainless ball bearings. This play could lead to premature failure of the bearings by inducing slight axial loads, but further testing is needed to confirm this.

5.5.2. Tensioner Arm Design

The tensioning system maintains pressure between the drive wheel and mooring line to prevent slippage, while also accommodating fluctuating tension loads from waves. The initial design featured a tension arm that pivoted around a hinge with a polytetrafluoroethylene (PTFE) polyamide bushing because it did not need sealing. However, this approach was not pursued further after testing revealed that the bushings were only PTFE coated, making them susceptible to corrosion if the coating was displaced. Instead, the entire hinge was constructed from acetal, which serves as a bushing material. This reduced the number of components and manufacturing processes required, while also making the system lighter and more cost-effective than the original hinge design. The original design was intended to be made from 316 SS. The lever arm and supporting pins were made from 316 SS to minimize deflections under load and to ensure corrosion resistance. A 316 SS torsion spring was used to maintain tension on the mooring line. This type of spring was selected over compression or tension springs due to its compact design and reduced likelihood of catching debris in the water.

The required spring torque was calculated by determining the minimum force needed for the drive wheel to grip the cable using the test setup depicted in the left image of Figure 11. Guide wheel positions were adjusted while the mooring line was loaded to the minimum expected tension force of 100 N. Calculations in Appendix C established that the minimum gripping force is 6 N. The system incorporates a safety factor of four to ensure that the drive wheel consistently has adequate traction. The completed hinge system is shown in Figure 12.

Figure 11: Pulley test setup (left) and 3D-printed tensioner prototype (right).

Figure 12: Completed tension arm design.

5.5.3. Drive Train Design

The crawling mechanism uses a drive wheel to propel the profiler along the mooring line. This wheel is driven by a motor. The drive system ensures the motor remains waterproof, isolates it from radial and axial loading, minimises friction, and has a horizontal profile to reduce drag and the risk of damage.

5.5.3.1. Sealing

Significant time was spent developing a system that efficiently transmitted power to the wheel without exposing the motor to salt water. Magnetic couplers were initially considered but discarded due to complexity and unreliability. A shaft driven solution was chosen instead. Existing shaft seals for underwater applications were researched by contacting Blue Trail Engineering. In their experience, X-profile O-rings are the best solution for sealing motor shafts in marine environments. They use these seals in their commercial underwater servo motors. They specified that surface finish and concentricity of the shaft are the most important factors to consider when designing a shaft for X-profile O-rings and any deflections must be kept to a minimum. A sample of the deflection calculations done for different shaft bearing and seal arrangements is in Appendix D. Standard O-rings are used as face seals for all other sealing surfaces involved in the assembly.

A cross-section of the drive system design can be seen in Figure 13. The main enclosure of the motor is made from acetal to reduce the weight of the assembly. This is paired with a stainless-steel face plate which components are mounted to. The interface is sealed by a rectangular groove O-ring. The bearing housing and motor are mounted to the plate using threaded blind holes, so the enclosure is waterproof. The plate is secured to the acetal main motor housing with bolts on an external flange outside the seal. The shaft bearing housing is machined from stainless steel for its excellent corrosion resistance and high strength. The housing was designed for high performance while being easy to assemble.

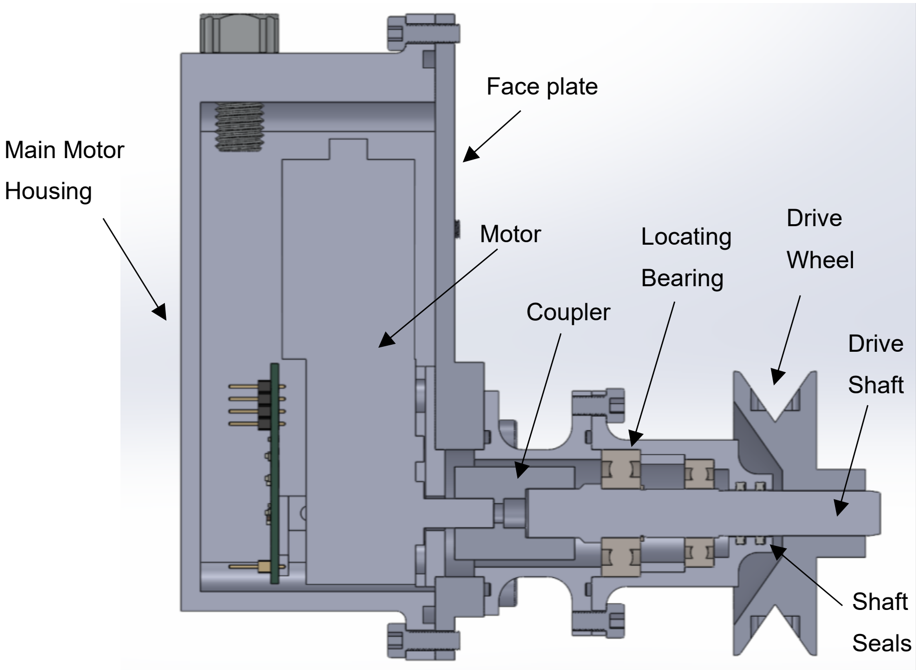

Figure 13: Cross section of the sealed motor housing and bearing arrangement with main components labelled.

5.5.3.2. Force Transmission

The bearing arrangement shown in Figure 13 was designed to isolate the motor from any radial and axial loads exerted by the tensioning system on the drive wheel. It also allows the shaft to rotate freely while ensuring concentricity with the O-ring seals. Placing the bearing in the motor enclosure extends its life by protecting it from saltwater. Both bearings are behind the shaft seals. The left-most bearing withstands axial loads. A shoulder and circlip constrain the inner race of the bearing on the shaft.

5.5.3.3. Profile

The drive system's profile determines its underwater drag and resistance against collisions. The first prototype used an axial motor that protruded from the chassis which the client highlighted as a point of failure. The issue was addressed with a redesign to incorporate a right-angled drive motor which reduced the axial length. Additionally, a redesign of the housing allowed the bearings which originally took the load from the wheel to be combined with the bearings supporting the shaft. This allowed an additional shaft and coupler to be removed. Furthermore, a hollowed-out wheel design was developed to make the design more compact. Iterations of the drive-train design can be seen in Appendix E.

5.5.3.4. Prototyping, construction and testing

To ensure that the chosen design was sufficient a functional prototype of the bearing and motor housing has been created. This allows for the design to be verified before costs associated with the machining are used. The prototype implemented O-ring seals on the shaft and sealing faces combined with bearings in the housing. An exploded view of the prototype motor housing can be seen in Figure 14.

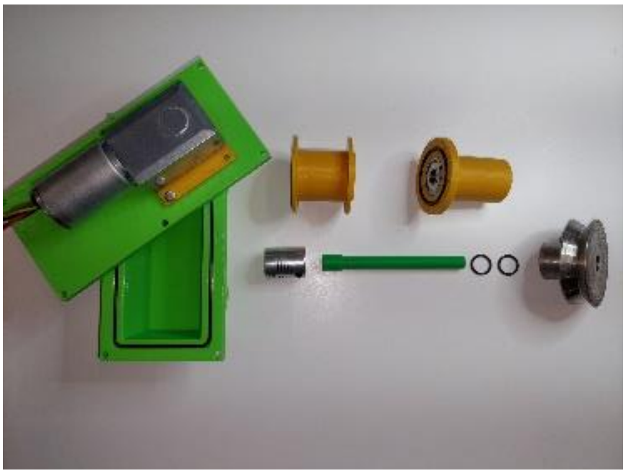

Figure 14: Exploded view of functional drive system prototype.

Once the prototype had been successfully constructed the design was machined from the final materials. Drawings were created for the Universities machinists. These drawings included bearing fits as well as shaft and O-ring tolerances to allow for watertight assembly. An image of the completed housing can be seen in Figure 15. Due to the considerations taken with regards to ensuring the assembly was watertight, assembly instructions can be seen in Appendix F.

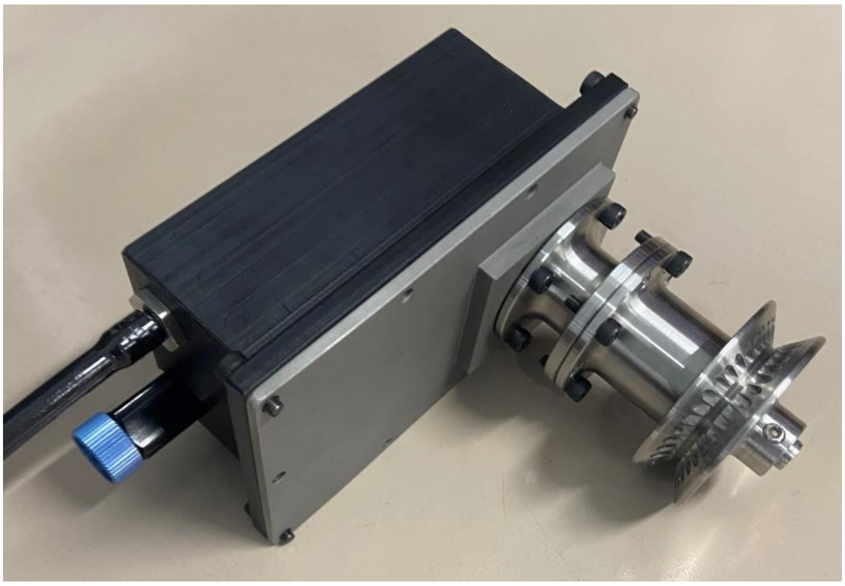

Figure 15: Completed motor design.

Following the construction of the housing testing of the sealing and drive properties was necessary. Testing of the sealing capabilities was done through a vacuum test. Creating a -1.0-bar internal vacuum inside the motor housing approximates an environment ten metres below the sea level. The housing maintained this vacuum for a period of 5 minutes which proved it sealed correctly. Radial forces were applied to the shaft during the test to determine whether deflection of the shaft compromised the shaft seals. This test set up can be seen Figure 16. The motor submerged for an underwater torque test can be seen on the right of Figure 16.

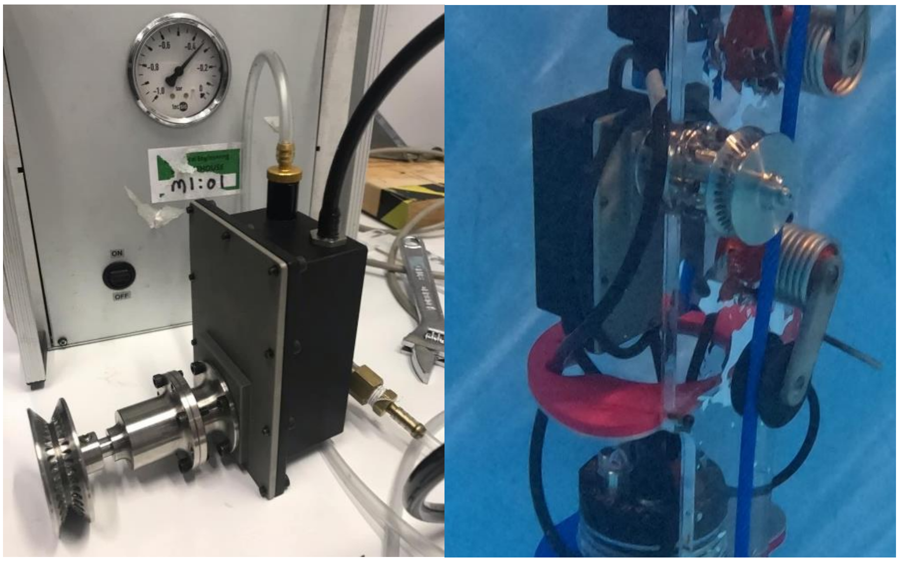

Figure 16: Motor housing vacuum test (left) and underwater drive test (right).

5.6. Charging Module Achievements

Niobium connectors, inductive charging, and mechanical airlocks were considered as potential methods for underwater charging (M16 Project Team, 2024a). Airlocks were considered too complicated and use of niobium for underwater charging was thought to be patented by Northrop Grumman. This left inductive charging as the only viable option. Subsequent research later in the year revealed that Northrop Grumman's patent was not filed in New Zealand, allowing niobium to be used in commercial work. Niobium and inductive charging were developed as two separate modules per Cawthron's request. These modules have been designed to be modular, so the niobium charging, and the inductive charging systems are interchangeable.

Experimental coil arrangements for charging autonomous vehicles are an area of ongoing research (M16 Project Team, 2024a; 2024b). A planar arrangement was selected, as these have seen commercial success in wireless phone chargers and off-the-shelf components already exist. A set of premade-coils and power electronics was purchased from DFRobot (SKU: DFR0712) and tested (Figure 17) to characterise their performance. They were then deployed on the profiler (Figure 18). The result of this testing is in Appendix G. The receiving coil is mounted on top of the profiler, and mates with the transmitting coil mounted on the buoy. To prevent biofouling on the coils, copper-impregnated PLA is used on the mating surface which releases copper ions, killing microorganisms attempting to grow on the surfaces. This ensures that biofouling will not interrupt the charging process over a multiple-month deployment. A detailed design of the completed modules is in Appendix H.

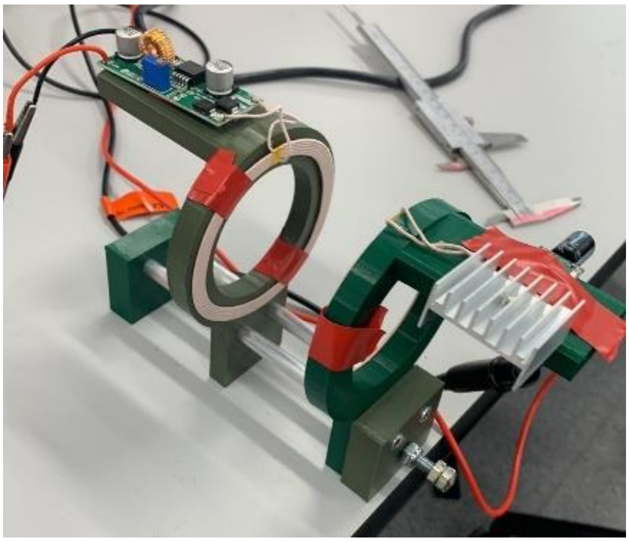

Figure 17: Inductive charger test stand.

Figure 18: Completed inductive charging modules.

Due to Northrop Grumman's patent, systems for underwater niobium charging are not commercially available, requiring a novel connector design to be developed. This design had several requirements. It had to be able to transmit power and communicate between the buoy and the profiler. It had to be able to form connections underwater in swells, prevent seizing or jamming in any orientation, and be self-locating to account for any misalignment which may come from ocean swells or the orientation of the profiler relative to the buoy.

A design was developed to meet these requirements (Figure 19). The plugs are spring-loaded to account for misalignment and introduce compliance to prevent damage. The plugs have tapered tips to slide into their sockets. The sockets have curved profiles align the plugs as they slide across the surface. To prevent potentially hazardous misconnections, the sockets and plugs are arranged asymmetrically, so they can either all come into contact correctly, or only one connection will be formed. As current flows in loops, at least two connections would be required to create a short-circuit or provide too much voltage to sensitive pins. This means that if the profiler mates with the buoy in the wrong orientation, it will not transmit data or recharge, which is preferable to damaging the components.

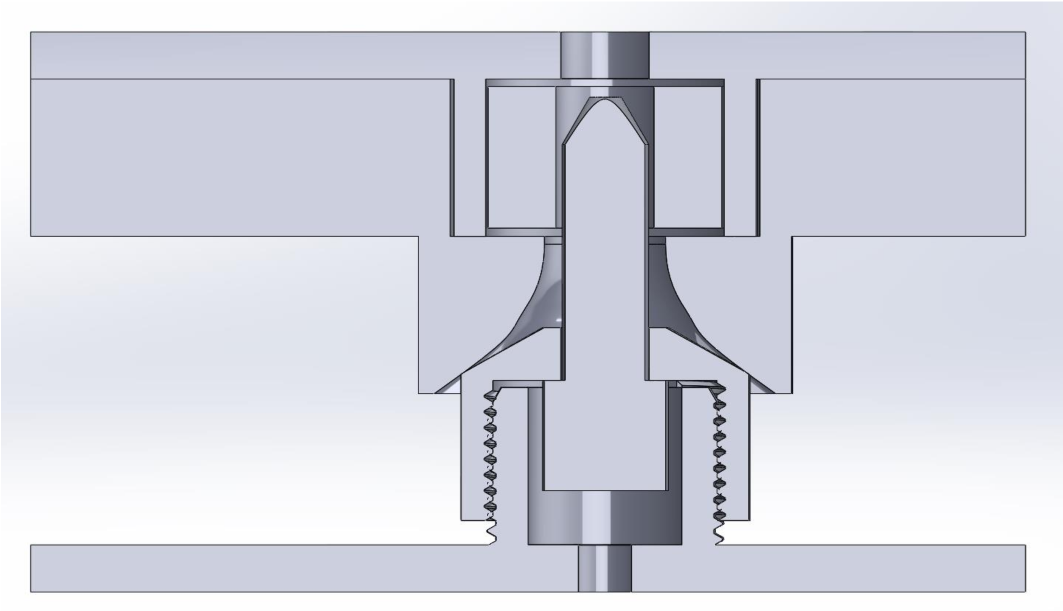

Figure 19: Niobium connector prototype.

The niobium plugs and sockets were manufactured and tested, with a measured resistance of 0.8 Ω. Figure 20 shows the final niobium connections.

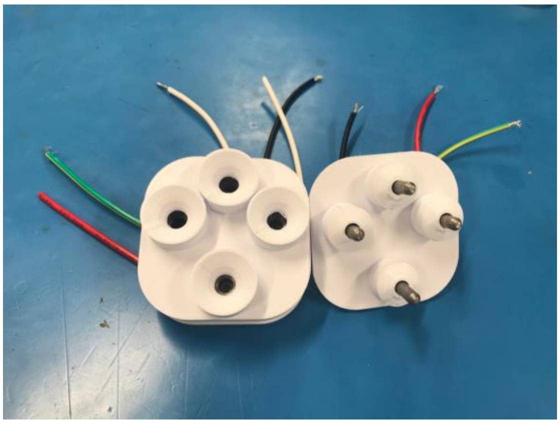

Figure 20: The final niobium charging prototype.

The inductive coils are sealed in an epoxy plate, which seals against a pressure vessel containing the power electronics as shown in Figure 18. This configuration can successfully transmit up to 8 W of power. It has approximately 85% efficiency, which may be further improved with the addition of a magnetic core.

5.7. Wireless Communications Achievements

The profiler and buoy must communicate with each other. As with the charging module, a wireless solution was considered alongside the niobium connector. The three main methods for wireless underwater communication are acoustic transmission, optical transmission, or electromagnetic transmission. Each method has drawbacks. Acoustic transmission is slower than electromagnetic transmission and can have negative effects on surrounding sea life. Optical transmission dependant on the clarity of the water it is used in. Water clarity is weather dependent, and passing storms could mean that the profiler is unable to transmit data for hours to days. Low frequency (< 30 kHz) electromagnetic transmission frequencies are restricted for military submarine use and free bandwidths which consumer electronics use for Wi-Fi and Bluetooth have wavelengths which are heavily attenuated by water molecules.

Due to the prevalence of 2.4 GHz devices, it was decided to test underwater communications with a pair of 2.4 GHz radios in salt water. After a successful set of tests with ranges up to 1.5 m, it was decided that 2.4 GHz radio could be used for reliable and fast underwater communications, as it was expected that the radios would be placed within 500 mm of each other. 2.4 GHz radios were preferred due to the availability of commercial solutions. It was decided to use the Bluetooth 5.0 protocol for more robust communication and potential support for other devices in the future.

The initial design housed the radio inside the CawPy's tube, but this was rejected in favour of putting the radios in their own housings to minimise the required transmission range. The current design has the radios installed inside the inductive charging housings on the custom PCB in Appendix I. There were originally concerns that the inductive charger would interfere with the radio communications, but testing has been completed to ensure the radio signal is not affected by the inductor's magnetic field.

The final design uses ESP32s for Bluetooth communications. This integrates with the wireless charging module, reducing the number of components needed. It can achieve 100 mm range with one module fully submerged and the other at the surface (Figure 21). This is theoretically sufficient but a test with both modules submerged is yet to be conducted. These modules meet all the clients' requirements, with the Bluetooth third party support exceeding the original specifications.

Figure 21: Testing surface-underwater radio communication range.

5.8. Software Achievements

Embedded software is needed to measure sensors and control motors on the robot. Cawthron specified that all code must be written in Micropython, as that is what they use in-house. Initially, a real time operating system (RTOS), synchronous, and asynchronous task scheduler software architectures were considered. The most commonly available RTOS is FreeRTOS, which is supported on Cawthron's hardware, but will not interface with Micropython without substantial C coding which the client would not be able to maintain. Synchronous task schedulers cannot operate as quickly as asynchronous schedulers but are significantly easier to implement. For this reason, initial versions were completed as synchronous models. The final version is asynchronous because Bluetooth is an asynchronous standard and it was easier to include in an asynchronous scheduler.

Bluetooth relies on a series of standards published by the Bluetooth special interest group (SiG). After consulting with Cawthron, the decision to make the robot Bluetooth compliant was made with future compatibility upgrades in mind. The standards used for reporting sensor measurements, battery levels and motor parameters are detailed in Appendix J.

The profiler's CawPy communicates with the motor and wireless modules over the RS232 protocol. Each module broadcasts their relevant parameters periodically as a formatted JSON string. The Micropython JSON library allows for automatic encoding and decoding of JSON strings into dictionaries which makes this message format easy to implement and scale. It is less efficient than sending raw byte arrays, but the robot is not operating fast enough for this to be an issue.

At a high level, the robot follows the flow chart shown in Figure 22. Programming for the buoy is out of scope. Lower-level functionality for during dive missions has been developed with a focus on easy expansion for future upgrades. The robot is expected to upload samples to the buoy via the file transfer protocol L2CAP, however this has not yet been implemented. Whilst battery status is available via Bluetooth, further work is required to develop a model for calculating recharge time for sleeping.

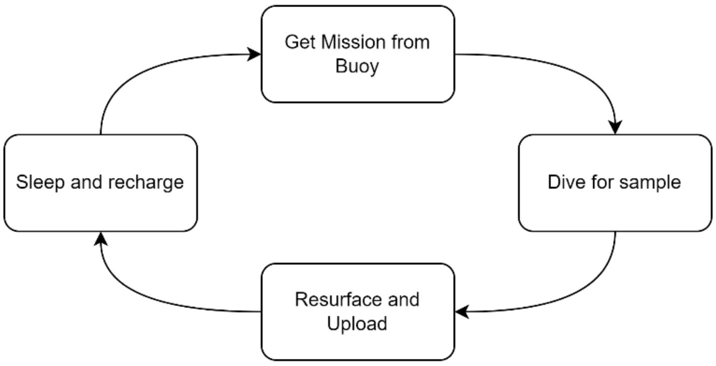

Figure 22: High-level operation flowchart.

The JSON communication format makes it easy to add additional modules to the robot. If each module complies with a standard meta-data format i.e 'sensor type', 'units', 'precision', 'data' then any sensor, motor, or other peripheral can be connected without reprogramming the CawPy. This is a valuable feature for commercial products as clients do not need to reprogram their own systems or replace multiple components each time they wish the change the sensor payload or upgrade a component. Cawthron already has a standard format for their other products which could be adapted for this purpose.

Overall, the software meets the Cawthron's need and implements all functionality required for demonstration and proof of concept. Further work is required before autonomous deployments can be done.

5.9. Motor Selection and Control Achievements

A critical requirement for this project is the minimum rate of one sampling cycle per hour. The underwater environment means the motor needs to have a torque rating of around 0.1 Nm to overcome drag and buoyancy forces, along with any obstacle which may grow on the mooring line. To avoid excessive energy consumption, the motor is required to operate at low rotational speeds while having a small form factor. Additional requirements are detailed in Appendix K.

The motor selection process is detailed in Appendix L and the chosen motor is a brushless DC motor. The custom breadboard controller shown in Figure 23 was made to experimentally confirm the motor performance parameters specified by the manufacturer. These tests confirmed that the motor can supply the necessary torque and does not draw more than 3 amps of current, which could damage the batteries.

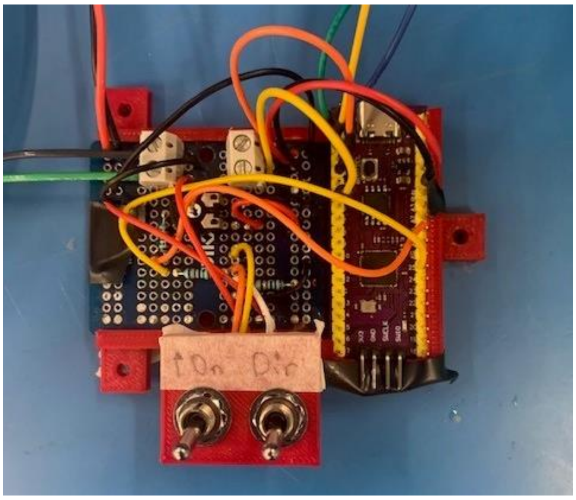

Figure 23: Manual motor controller.

Unlike a brushed DC motor, brushless motors need additional wires for speed and direction control, so a custom interface was designed to allow the CawPy to control the motor. The CawPy also communicates through its external ports using the RS-232 protocol, so an interpreter is required to convert between the speed instructions sent from the CawPy and the PWM that the motor expects. A PCB was designed to perform both functions.

The interface design is detailed in Appendix M and was implemented using the same PCB layout as the communication and inductive charging module. The PCB is installed in the motor housing. This motor controller can also regulate the motor's power consumption via speed control and has an encoder for closed loop motor control. The PCB mounted on the motor is shown in Figure 24.

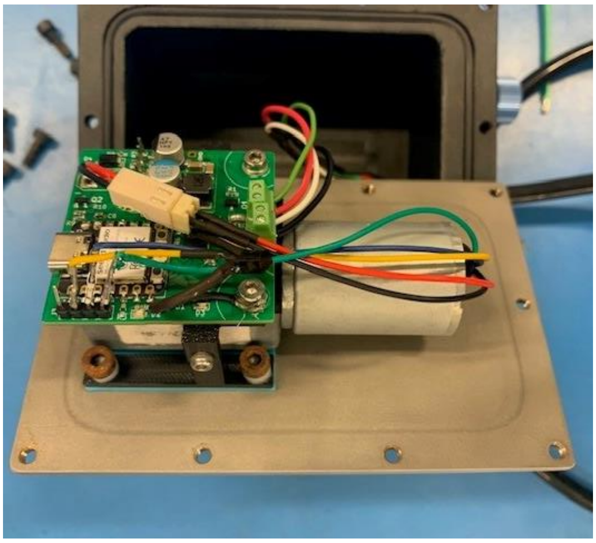

Figure 24: Motor PCB mounted on the motor.

The automatic motor controller was reworked to account for a motor peculiarity. The direction signal for the motor requires a high-impedance signal to go backwards, and a digital low to go forwards. The PCB sends a digital high signal to go backwards, which was incompatible with the motor. This problem was rectified by inserting a MOSFET transistor with the gate connected to the drive signal, the source connected to ground, and the drain connected to the motor drive pin. This converted the digital signals into the high-impedance and low signals the motor was expecting, allowing full controllability of the motor.

5.10. Docking Station Achievements

To recharge the profiler, a docking station is suspended beneath the surface buoy. This prevents the rough surface water from destabilizing the profiler while docking. This docking station is where the niobium or inductive charging is implemented and where the profiler will transmit data to the buoy. The docking station is clamped to a section of the mooring line approximately 1 meter below the buoy. This station needs to be designed to fit through the 260 mm hole in the centre of the buoy. The docking station final prototype can be seen in Figure 25 at two stages of the docking process.

As this part of the project relied heavily on the final geometry of the profiler it still has work to be completed. An initial design was created in term 1 however the design was halted due to multiple unknowns about the shape and charging requirements of the profiler. Upon the completion of the profiler prototype in term 4 the design was further refined, and a prototype was created from 3D printed materials to test the function of the design. The test was a success, and the docking cage rotated to align the inductive coils.

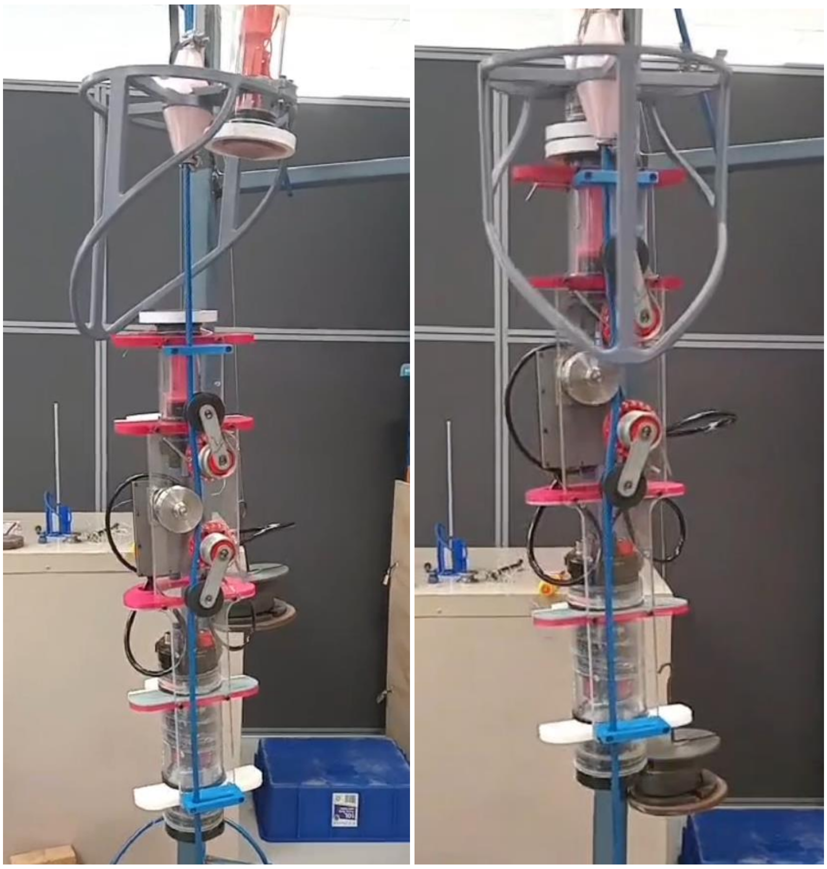

Figure 25: The docking station rotating to align with the profiler.

5.10.1. Orientation Mechanism

When the profiler finishes its temperature profile the cage will rotate around the cable to align itself, ensuring the profiler reaches the apex of the cage. Weak magnets will be installed at the apex of the cage to help secure the profiler in place during charging. They must be weak enough for the motor to overcome them once charging is finished. The most suitable material for the cage is stainless steel because it is corrosion resistant, has good material properties and is non-magnetic. This means that it will not interfere with the locating magnets or inductive charging.

5.10.2. Charging Implementation

The docking station has been designed to be compatible with both niobium and inductive charging methods. The charging components are in separate 2" pressure housings on both the profiler and the docking cage to ensure interchangeability with the niobium alternative. The 2" pressure housing is mounted at the apex of the docking cage, allowing it to mate with the corresponding receiver on the profiler. The 2" pressure housings can be seen in Figure 25 with the white ring visible which houses the inductive coil.

5.11. Sensor Achievements

5.11.1. Temperature Sensor

Cawthron selected a temperature sensor as the highest priority sensor to deploy on the profiler for its importance to commercial mussel farmers. Unfortunately, no commercially available temperature sensor was rated requirements for a 12-month continuous deployment. Consequently, the team to develop a waterproof pressure housing for an existing sensor circuit to protect it from damage without impacting response time. An innovative enclosure was developed (Figure 26) using vacuum forming to protect the sensor. The detailed design process is in Appendix N. The sensor enclosure is sealed using epoxy and has been installed on the CawPy. A driver and test files were written in Micropython to confirm it performed as expected. The temperature response was measured in warm water and met the requirements by detecting a 10 degree change in under five seconds.

Figure 26: Completed temperature sensor.

5.11.2. Pressure Sensors

The pressure sensor is crucial for determining the profiler's depth when taking profiles. The pressure sensor used (Figure 27) is a commercial product from BlueRobotics. The sensor has a theoretical accuracy of ± 30 mm but this needs to be experimentally validated.

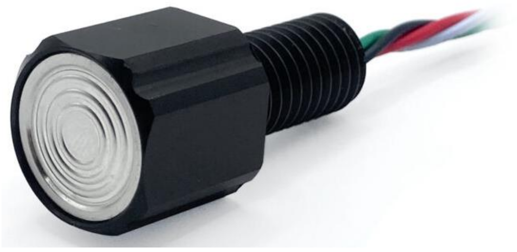

Figure 27: Bar100 Pressure Sensor

6. Schedule

The final project Gantt Chart is shown in Figure 28.

i	Task Name	Duration	Start	Finish	Predecessors
✓	**Project Set-up**	**24 days**	**Mon 26/02/24**	**Thu 28/03/24**	
✓	Meet with client and confirm project goals	5 days	Mon 26/02/24	Fri 1/03/24	
✓	Draft Proposal	11 days	Mon 4/03/24	Mon 18/03/24	2
✓	Final proposal	7 days	Wed 20/03/24	Thu 28/03/24	3,5
✓	Confirming specifications	11.68 days	Mon 4/03/24	Mon 18/03/24	
✓	**Docking Mechanism**	**144 days**	**Mon 26/02/24**	**Thu 12/09/24**	
✓	Research	14 days	Mon 26/02/24	Thu 14/03/24	
✓	Concepts	14 days	Sat 16/03/24	Wed 3/04/24	
✓	Detailed design	28 days	Thu 1/08/24	Mon 9/09/24	
✓	Visualisation modelling (small scale models etc.)	28 days	Sun 4/08/24	Tue 10/09/24	
✓	Physical model key components	25 days	Sat 10/08/24	Thu 12/09/24	
✓	**Crawling Mechanism**	**122 days**	**Mon 26/02/24**	**Tue 13/08/24**	
✓	Research	14 days	Mon 26/02/24	Thu 14/03/24	
✓	Concepts	36 days	Sat 16/03/24	Fri 3/05/24	
✓	Detailed design	33 days	Fri 14/06/24	Tue 30/07/24	
✓	Visualisation modelling (small scale models etc.)	54 days	Fri 5/04/24	Wed 19/06/24	
✓	Physical model key components	28 days	Fri 5/07/24	Tue 13/08/24	
✓	**Power Electronics**	**100 days**	**Mon 26/02/24**	**Fri 12/07/24**	
✓	Research	14 days	Mon 26/02/24	Thu 14/03/24	
✓	Concepts	14 days	Sat 16/03/24	Wed 3/04/24	
✓	Design	48 days	Fri 29/03/24	Tue 4/06/24	
✓	Component Testing	46 days	Fri 10/05/24	Fri 12/07/24	
✓	**Embedded Systems**	**100 days**	**Mon 26/02/24**	**Fri 12/07/24**	
✓	Research	14 days	Mon 26/02/24	Thu 14/03/24	
✓	Concepts	14 days	Sat 16/03/24	Wed 3/04/24	
✓	Design	50 days	Fri 29/03/24	Thu 6/06/24	
✓	Component Testing	46 days	Sat 11/05/24	Fri 12/07/24	
✓	**Construction and Testing**	**65 days**	**Fri 12/07/24**	**Thu 10/10/24**	
✓	Prototype construction	56 days	Fri 12/07/24	Fri 27/09/24	
✓	**Testing**	**15 days**	**Fri 20/09/24**	**Thu 10/10/24**	
✓	Vacuum test	1 day	Fri 20/09/24	Fri 20/09/24	
✓	Indoor vertical drive test	1 day	Fri 27/09/24	Fri 27/09/24	
✓	Pool test	1 day	Thu 10/10/24	Thu 10/10/24	
✓	Orientation mechanism test	1 day	Thu 10/10/24	Thu 10/10/24	
✓	**Presentation**	1 day	Fri 11/10/24	Fri 11/10/24	
	Handover preparation	**22 days**	**Fri 11/10/24**	**Mon 11/11/24**	
	Assembly instructions	22 days	Fri 11/10/24	Sun 10/11/24	
	Documentation	22 days	Fri 11/10/24	Sun 10/11/24	
	CAD packaging	22 days	Fri 11/10/24	Sun 10/11/24	
	Handover	1 day	Mon 11/11/24	Mon 11/11/24	

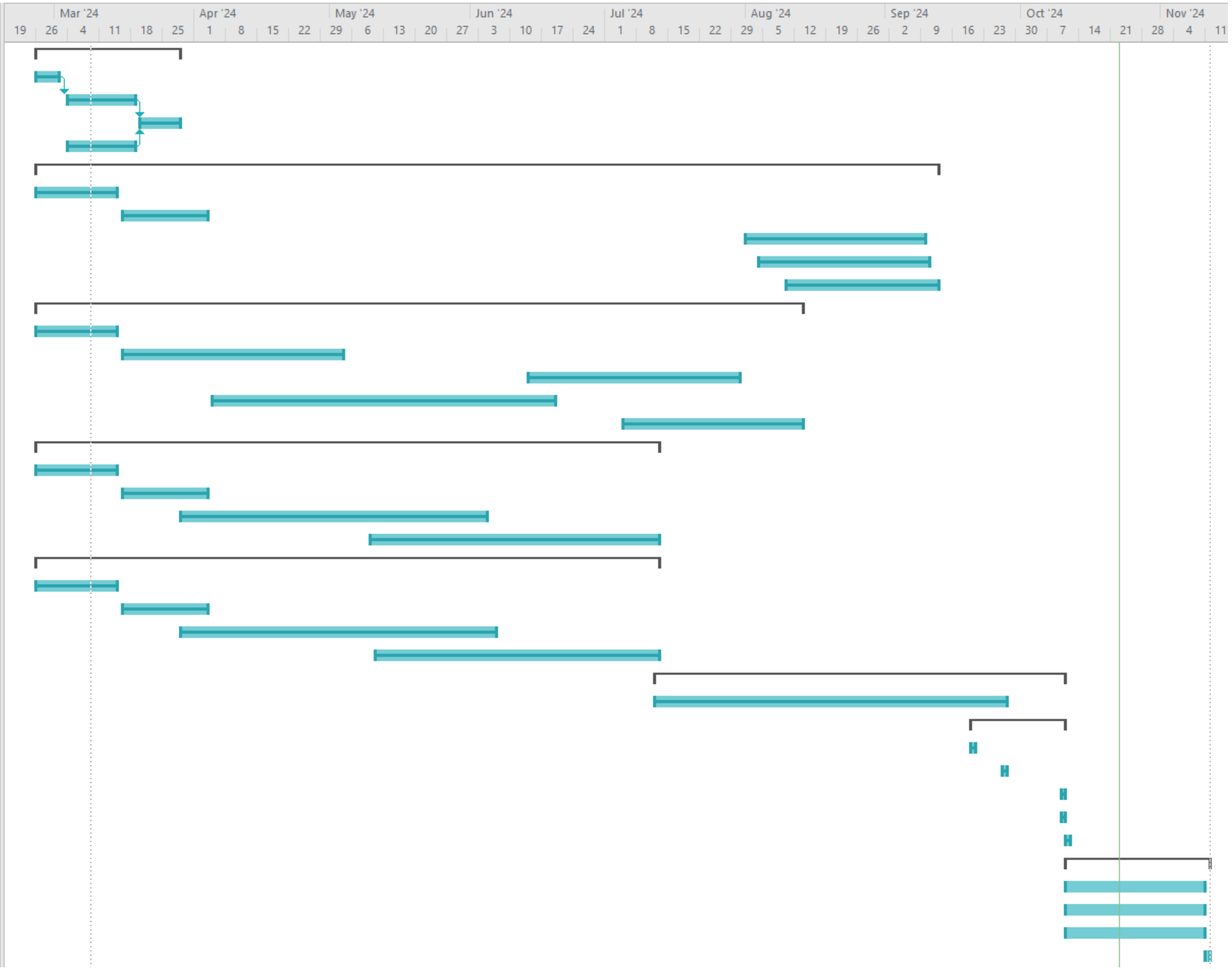

Figure 28: Updated Gannt Chart.

6.1. Commentary on Gantt Chart Changes

The Gantt chart has not changed significantly since the mid-year report (M16 Project Team, 2024b). Extra tasks have been added, such as the handover preparation, which were omitted from the mid-year report. Additionally, the Gantt Chart described in Figure 16 is more explicit about which tests have been completed and when. The mid-year report contained very few details about the testing as the tests had not been designed at that stage.

Key dates for producing and testing the submodules were all met as specified in the mid-year report. This was because the designs for all submodules had gone through multiple prototypes by the time the mid-year report was published, and there was a good understanding of changes which needed to be made and why for each module. The mid-year report Gantt chart included time for revising each submodule once more, and then the time required to produce the completed modules. This was a straightforward process, and after manufacturing the designs, only a few submodules needed further revisions. The major components which needed construction, which included the motor housing, drive train, tensioner arms, and the inductive coil housings, did not require any revisions after their manufacture. As these major components had worked as intended due to their extensive prototyping, they were produced on time as expected in the Gantt Chart.

7. Updated Budget

7.1. Financial Tracking

Table 3 describes an updated budget including current and projected expenditure.

Table 3: Project budget with spending tracked.

Crawling Mechanism					
Item	**Description**	**Rationale**	**Cost**	**Provided**	**Spent to date**
Bearings	SS Sealed for life	For rotation of wheels	$70.00	No	$70.00
Acetal Bar Stock	For machining idler wheels and hinges	Necessary for saltwater operation	$50.00	Yes	$50.00
Stainless steel stock	For machining motor housing and tensioner	Necessary for saltwater operation	$80.00	No	$80.00
Motor	DC brushless motor	Drives profiler along line	$290.00	No	$290.00
Acetal stock	For making chassis	Light and cheap	$127.00	No	$127.00
Off-site acetyl machining	For making chassis	Insufficient UC facilities	$116.00	No	$116.00
Stainless steel springs	316 stainless torsion springs	Provides tension for mechanism	$170.00	No	$170.00
Circlips	316 SS Circlips	For shaft assembly of motor housing and guide wheels	$16.50	No	$16.50
O-rings	Multiple O-ring sizes	For sealing of motor housing and guide wheels	$46.34	No	$46.34
Docking Mechanism					
Item	**Description**	**Rationale**	**Cost**	**Provided**	**Spent to date**
Magnets	Neodymium coin magnets	For keeping the profiler secured whilst docked	$40.00	No	$0
Electronics					
Item	**Description**	**Rationale**	**Cost**	**Provided**	**Spent to date**
Pressure Sensor	Blue Robotics Bar100	Rated for 12 months continuous deployment	$542.00	Yes	$542.00
Temperature Sensor	Blue Robotics Fast Response	Very accurate with fast response time. From a trusted supplier	$130.00	Yes	$130.00
CawPy Mini 2.0	Cawthron's Proprietary microcontroller	Ensures device will be compatible with the rest of Cawthron's products	$1,000.00	Yes	$1,000.00
Seeed Xiao Microcontollers	Wifi-enabled microcontrollers	For communications	$27.00	No	$34.00
Inductive Charger	DFRobot 12V 3A wireless charger	Removes the need to develop a custom inductive charger	$82.00	No	$82.00
Batteries	4 x 3.8Ah 26650 LiFePO4	Used to power profiler	$60.00	Yes	$60.00
Voltage Regulators	12V-5V DC buck converters	Used to power Seeed Xiao microcontrollers from CawPy mini	$37.50	No	$37.50

UART converters	MAX3232 5V to 3V3 uart transceiver	Used to communicate between the Seeed Xiao and the CawPy mini	$8.00	No	$8.00
Custom PCB Order	Custom PCBs from JLC PCB	Acts as mounting boards for the Seeed Xiao and MAX3232s.	$22.00	No	$22.00
Pressure Vessel Components					
Item	**Description**	**Rationale**	**Cost**	**Provided**	**Spent to date**
Pressure Vessels	2 x 2" 150mm cast acrylic. Blue Robotics	For housing the inductive charger and radio	$92.00	Yes	$92.00
Cap flanges	2 x 2" machined acetal flanges	For mounting the inductive charging coils to their pressure vessels	$98.00	Yes	$98.00
End Caps	2 x 2" 2-hole machined acetal end caps	For sealing the inductive pressure vessels	$103.00	Yes	$103.00
Spare O-Rings	Spare 2" O-rings from blue robotics	Used for sealing the 2" pressure housings	$6.00	Yes	$6.00
6 Pin Bulkheads	6 Pin cobalt Connectors	Mounted to the CawPy housing	$170.00	Yes	$170.00
6 Pin cables	2 x 6 pin 1m cables. BlueTrail Engineering	Connects the wireless module to the CawPy	$180.00	Yes	$180.00
Pressure Relief Valves	Blue Robotics	Used to vent gas in pressure vessels	$138.00	Yes	$138.00
Miscellaneous					
Item	**Description**	**Rationale**	**Cost**	**Provided**	**Spent to date**
Mooring Line	8mm Plastic jacketed mooring line	Used for design testing. Same as wire planned for final deployment	$100.00	Yes	$100.00
Niobium	Niobium round stock	Used to machine pegs for niobium connector module	$200.00	No	$200.00
Travel to Nelson	85c/km ute cost	Visit Cawthron	$1,100.00	No	$1,100.00
Accommodation in Nelson	$100pp for four	Staying overnight	$400.00	No	$400.00
		Total costs:	**$5501.34**		**$5,468.34**

Of the $5,468.34 which was spent on this project, when the costs of the items that were provided by Cawthron were removed, the project has spent $2799.34.

7.2. Changes from Mid-Year Report Budget

There are only a few minor changes since the mid-year report budget. The first was the exchange of stainless steel for acetal for the chassis, which was considerably cheaper. The second change is the inclusion of a fourth Seeed Xiao to replace one which was damaged. This budget is $498.66 less than what was presented in the mid-year budget.

8. Implications

The project findings have been largely positive. All key subsystems have been designed, manufactured, and tested to various extents. However, there were requirements that have not yet been fully assessed. A list of tests which are yet to be completed is available below.

- A depth test to ensure that the profiler housings do not leak under 3 bar of pressure.
- A long-duration deployment to determine how long the profiler can function, and whether this is greater than the minimum six months established in the requirements.
- A battery life test, to ensure that the profiler can run for up to 72 hours off battery alone.
- Setting up a Bluetooth link with an existing Cawthron buoy to communicate observation data.

The team is confident that the design will pass each of the above tests, but there may be issues with the design which will only become apparent when tested to a greater degree. There are also some changes that the team would have liked to have made to the design, but ultimately ran out of time to implement them. These changes include:

- Reprogramming the power systems to allow for a lower current threshold to unlock faster battery charging.
- Adding more sensors, such as PAR and salinity sensors.
- Communicating using Bluetooth between two CawPys, one of which will be on the profiler and the other will act as a buoy.

Additionally, a mechanism for the docking cage to rotate around the cable needs to be tested. Concepts have been developed involving acetyl sliding surfaces to eliminate the need for bearings. As the cage is more likely to experience biofouling than the profiler due to its fixed depth, removing bearings allows for reliable rotation. Once the design has been finalised the cage will likely need to be constructed out of 6 mm 316 stainless steel tube. More work to ensure reliable charging while docked would also be beneficial. The inductive charger and niobium charging designs do not currently have a way to ensure that the correct proximity and orientation is maintained during charging. It is possible that relative motion between the profiler and the buoy could occur, leading to unreliable charging or communications.

The findings from this project have business implications for Cawthron. The Project has created a working robot which can carry sensor payloads, recharge itself, and transmit data to one of Cawthron's buoys. With further development it could become a commercially viable product. This will be the low-cost solution for coastal profiling which was Cawthron's main objective. In New Zealand, Cawthron will be able to use niobium connectors to charge and communicate efficiently, whereas any overseas deployments will have to use the wireless module due to Northrup Grumman's patent.

There will be social, environmental, and economic outcomes. From a social perspective, local councils and other organisations will be able to monitor ocean water quality at a considerably lower cost than existing means. This will make water quality monitoring more commercially viable. From an environmental perspective, it is hoped that increased monitoring will promote more effective solutions for protecting New Zealand's coastal waters. A few hundred micrograms of copper ions will be released into the water to clean mating surfaces on the docking mechanism during deployments. This is an industry standard for protecting ships. These ions will kill microorganisms which could cause biofouling in a local region immediately surrounding the mechanism but will not pose a threat to any larger wildlife at such a low concentration. Economically, this project has implications for New Zealand aquaculture industries. Mussel and fish farmers will be able to keep their stock in the optimal temperature region for growth and increased yield.

9. Future Issues and Limitations

While the wireless charger is capable of supplying 10W, when connected to the CawPy, its battery management chip limits the power to 1.8 W. This is because this chip uses a power regulation algorithm designed for solar panels. Further work to replace this algorithm with one suitable for the wireless charger is required.

The main power regulator has a minimum voltage threshold very near the battery minimum voltage. This means the power regulator shuts-down when the motor is under heavy load, effectively limiting the motor to approximately 30% of its maximum torque. An easy solution would be to increase the number of series battery circuits to increase the operating voltage.

The Bluetooth file transfer protocol (L2CAP) has not been implemented. This will be necessary for transferring bulk sensor readings in a timely manner. Further refactoring should be undertaken to improve the brains' ability to detect what is connected to its RS232 ports.

The deployment life of the crawling mechanism has not been verified. Components may wear unexpectedly in long deployments. These wearing components include the mooring line guides, as they are made from acetal which will be potentially damaged by the wear from the mooring line.

The guide wheel bearings will also wear as they experience varying loads and exposure to contaminants. The mooring line life is also a concern as the drive wheel "bites" and damages the line when it slips, which could occur as the buoy rises and falls at the water's surface. This may be resolved by deburring the slots in the V-groove of the wheel or using a harder plastic jacketed cable. The current guide design makes it challenging to attach the profiler to the mooring line as the fasteners are used for attaching multiple parts together. This could be resolved by changing the design to have more fasteners, although this will complicate the design and potentially make the part larger and more difficult to manufacture.

The chassis has limited adjustability and space for other components. This is because the profiler must fit through the centre of the buoy, which has a diameter of 260mm. Therefore, other components should be added along the robot's length. The chassis is also designed to accommodate a 190 mm diameter hydrodynamic body which has not been designed. This increases the robots drag underwater, limiting its efficiency. There is also potential for debris such as seaweed or fishing line to get caught on components.

10. Conclusion

In the face of a changing climate, it is increasingly important to develop an understanding of how ecosystems respond to environmental changes. The Cawthron Institute has commissioned an autonomous underwater profiler to carry sensors through a column of water to take measurements of interest, such as temperature. A set of requirements for the project were developed, and a design has been completed in accordance with those requirements. The profiler has been tested extensively and meets the initial project requirements. The project was broken down into submodules, which include a chassis design, a crawling mechanism, a charging module, a communications module, software development, motor control, a docking station, and a sensor payload. Each submodule has been tested individually to ensure that it performs as expected, and the submodules have been assembled into a complete prototype.

The project spent $2799.34, with Cawthron provided components costing an additional $2669.00 for a final cost of $5468.34. The project has several implications for Cawthron's business and future development of the profiler. The next steps should be to validate the robot's long term deployment requirements. There are also tasks which the team did not manage to complete but would recommend as the starting point for a future project. Ultimately, the project has produced a profiler prototype which is well on the way to meeting all the design requirements and functioning as desired by Cawthron, for a fraction of the price of existing commercial solutions.

11. Ethical Issues

No ethical considerations present after consulting with Cawthron's Mātauranga Māori advisors.

12. Contribution Statement

There are four members of the student team: Matthew King, Ben Stirling, Jacob Rhodes, and William Rawlins. Matthew has developed the profiler's embedded system including the niobium and inductive charging systems, motor control and temperature sensor enclosure. Ben also contributed to these systems in particular the radio communication, motor selection, administrative tasks, prototyping and testing. Jacob designed the drive train system, including the motor pressure vessel, housings containing the drive shaft, seals, coupling and bearings. Jacob has also developed the docking mechanism responsible for orientating the profiler during charging and data transfer. William developed the crawling mechanism design, researched options, created system models, and developed the drive wheel and chassis designs. The team supervisor, Shayne Gooch has been critical in providing expert advice with regards to mechanical design decisions and developing design concepts.

The team mentors Paul Barter and Shaun Graham (both from Cawthron) have been essential in providing insight into designing of the profiler, ordering materials and components. The university's lead mechanical technician Tony Doyle has provided help to manufacture parts in the machine shop and procure material for prototype designs. Julian Phillips has also assisted by providing the electrical and mechanical components for prototyping. Oscar Torres provided space and components for conducting tests. Scott Lloyd has helped the team source electrical components and repair surface mount electronics. Julian Murphy has also provided the team with components and equipment to test the system. The team would also like to acknowledge the contributions of Blue Trail Engineering in providing advice and guidance regarding underwater seals and magnetic couplers.

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14. Additional Information

A list of parts can be found in Table 4, which shows all the components for the subassemblies of the profiler.

Table 4. Parts List

Tensioner Assembly		
Part Description	**Material**	**QTY**
Hinge	Ø60mm Acetal	2
Hinge Pin	Ø18mm 316 SS	2
Tension Arm	5mm 316 SS Bar Stock	2
Bearing Pin	Ø18mm 316 SS	2
9x22x7 Ball Bearing	Acetal & 316 SS	2
Guide Wheel Body	Ø50mm Acetal	2
Guide Wheel Cap	Ø34mm Acetal	2
Guide (Bottom)	10mm Acetal Sheet	2
Guide (Top)	10mm Acetal Sheet	2
Custom Torsion Spring	316 SS Wire	2
Chassis Assembly		
Part Description	**Material**	**QTY**
Main Plate	10mm Acetal Sheet	1
Rib 1 (Base)	10mm Acetal Sheet	1
Rib 1 (Clamp)	10mm Acetal Sheet	1
Rib 2 (Base)	10mm Acetal Sheet	1
Rib 2 (Clamp)	10mm Acetal Sheet	1
Rib 3 (Base)	10mm Acetal Sheet	1
Rib 3 (Clamp)	10mm Acetal Sheet	1
Rib 4 (Base)	10mm Acetal Sheet	1
Rib 4 (Clamp)	10mm Acetal Sheet	1
Rib 5 (Base)	10mm Acetal Sheet	1
Rib 5 (Clamp)	10mm Acetal Sheet	1
Rod Brace 1	Ø12mm Acetal	4
Rod Brace 2	Ø12mm Acetal	1
Rod Brace 3	Ø12mm Acetal	2
Drive Train Assembly		
Part Description	**Material**	**QTY**
Brushless Motor, Right Angle Gearbox	12V, 92RPM.	1
Motor Housing	Acetal Stock	1
Mounting Lid	10mm 316 SS Plate	1
Plate for motor	PLA (3D Print)	1
Coupler Cover	Ø50mm 316 SS	1
Bearing Housing	Ø50mm 316 SS	1
Drive Wheel	Ø60mm 316 SS	1
Spacer	Ø25mm 316 SS	1
Drive Shaft	Ø10mm 316 SS	1
Single Row Deep Groove Sealed Ball Bearing	10mm ID x 26mm OD x 8mm, SS	1

Single Row Deep Groove Sealed Ball Bearing	10mm ID x 22mm OD x 6mm, SS	1
Flexible Coupling	Aluminium	1
Circlip	Ø10 mm SS	
Docking Mechanism		
Part Description	**Material**	**QTY**
Docking cage	6 mm SS plate, 6mm SS Tube	1
Charging module upper clamp	10 mm Acetyl Plate	1
Charging module lower clamp	10 mm Acetyl Plate	1
Cable clamp	SS Wire clamp	2
Acetyl sliding surface	4 mm Acetyl Plate	1
Charging Module Assembly & Electronics		
Part Description	**Material**	**QTY**
Blue Robotics 2” Locking Tube	Acrylic	2
Blue Robotics 2” Locking Flange	Acetal	2
Blue Robotics 2’’ End Cap	Acetal	2
CawPy Mini 2.0	-	1
Blue Robotics Bar 100 Pressure Sensor	-	1
Blue Robotics Fast Response Temperature Sensor	-	1
DFRobot 12V 3A wireless charger	-	2
4 x 3.8Ah 26650 LiFePO4 Batteries	-	4
12V-5V DC buck converters	-	-
MAX3232 5V to 3V3 UART transceiver	-	2
Custom PCB from JLC	-	2
6 Pin Bulkhead	-	2
6 Pin Cables (1m)	-	2
Niobium Pins	Ø8mm Niobium bar	4
Niobium Sockets	Ø16mm Niobium bar	4
Ø10mm Springs	Plastic	4
Socket Housing	PLA	1
Plug Housing	PLA	1
Miscellaneous		
Part Description	**Material**	**QTY**
M4 Cap Screws	316 SS	-
M10 Bulkhead Vent	Anodized Aluminium	-
X-Profile O-Ring	-	4
2” O-Rings	-	2
8mm Circlip	316 SS	5

15. Appendices

15.1. Appendix A – Mooring Guide Development

Figure 29 shows the initial mooring line guide designs and their iterations as problems associated with their geometric characteristics are addressed.

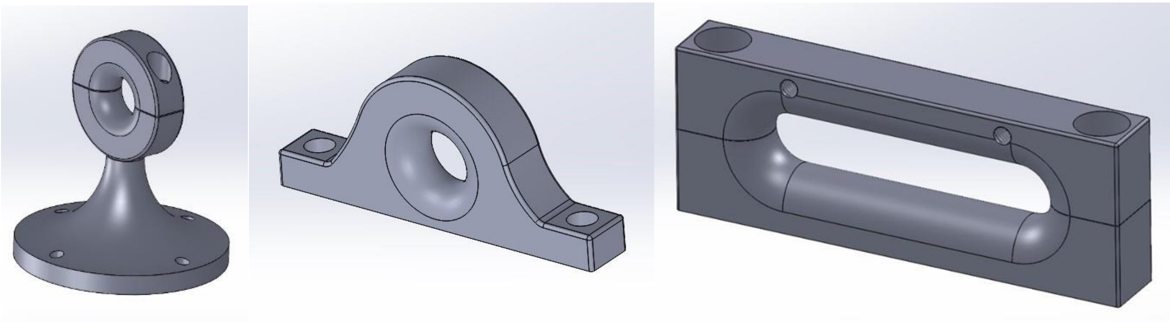


Figure 29: The mooring guides have undergone changes to improve reliability and performance.

15.2. Appendix B – Chassis and Crawling Mechanism Design Process

Initial concepts of the crawling mechanism involved various arrangements of wheels and pulleys, along with a claw/climbing mechanism and propellor thrusters. Some wheel and pulley arrangements are shown in Figure 30 but these were not pursued as they added unnecessary complexity and guide wheels are not suitable for loads parallel to the axis of rotation. Figure 31 shows a schematic of a claw-like mechanism and how it operates. This mechanism is not suitable as it can cause damage to the mooring line and the complex motion, and parts increase the likelihood of failure. Propellor/thruster systems were not explored further as they are inefficient at converting electrical energy to kinetic energy, which is critical for this application.

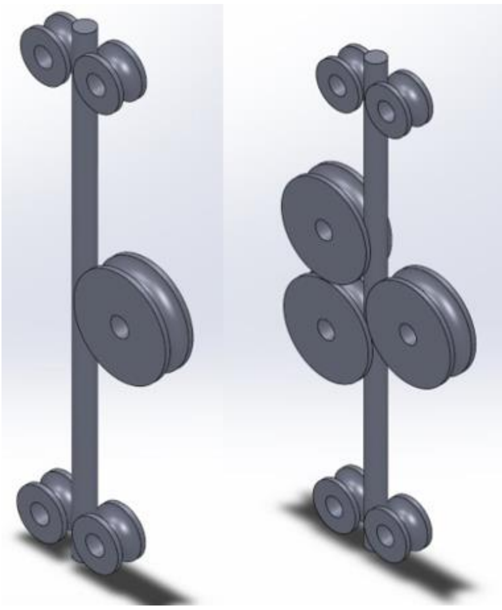


Figure 30: Discarded wheel and pulley arrangements. These were discarded as they were too complicated and more likely to fail than current designs.

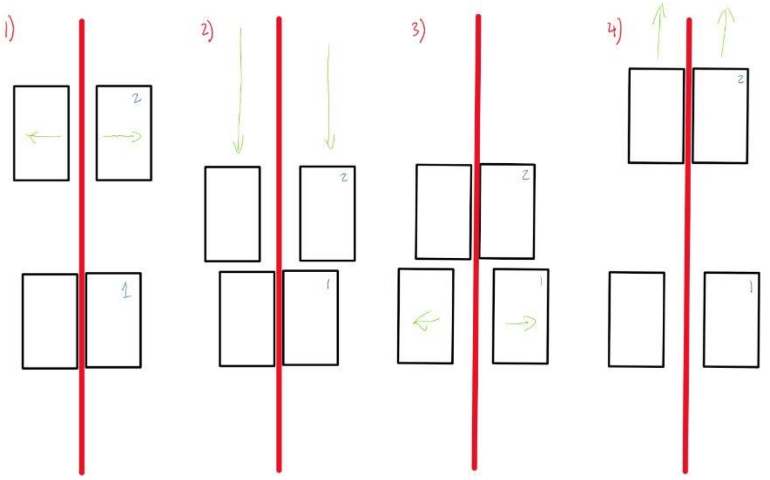


Figure 31: A depiction of how a claw mechanism would climb the cable.

Figure 32 shows the progression of the profiler chassis design and layout, where sheet metal was first implemented and replaced by a combination of bar and sheet metal. The latest iteration (not shown) utilizes Acetal support structures to reduce weight. The full sheet metal design (centre) was not pursued further because it has a large drag profile in the water and has limited strength properties in comparison to a bar and sheet system. The sheet and bar (right) were not pursued either even with better drag characteristics due to the associated weight material cost.

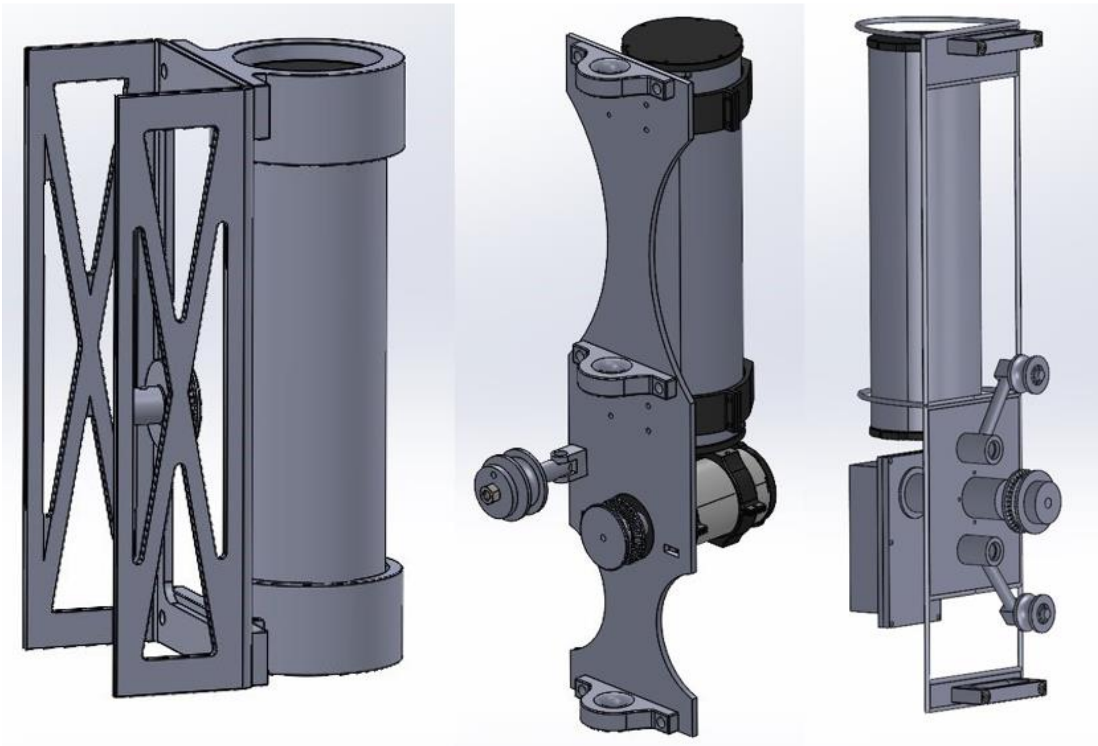
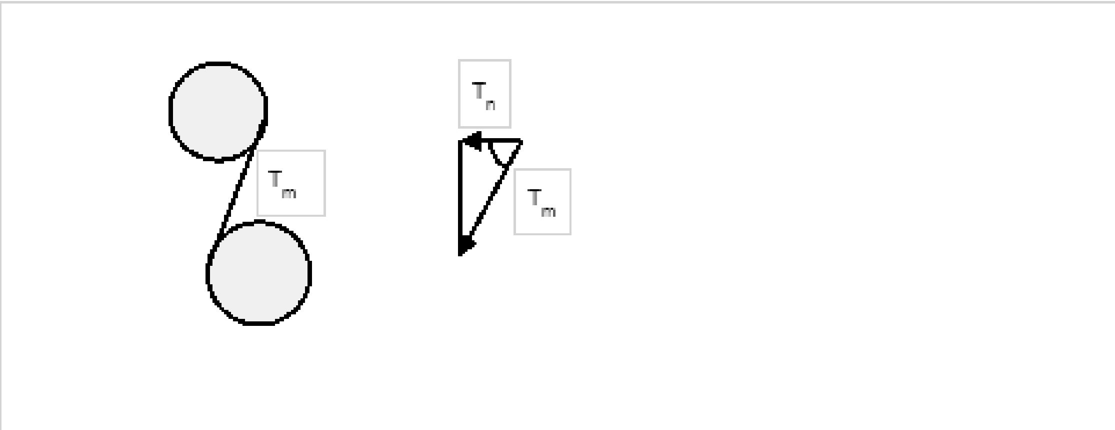


Figure 32: Past chassis design iterations

15.3. Appendix C – Profiler Force Calculations

Crawler Forces



The mooring line will have a 20kg weight attached at the end, which is suspended in the water column. This weight will move vertically with the swell of the ocean, meaning that a change in direction will result in a momentary loss of tension within the mooring line. It is assumed that the tension at this moment will be a minimum of 10kg.

Weight: $m = 10\ kg$

gravity: $g = 9.81\ \frac{m}{s^2}$

Mooring line Tension: $T_m = m * g = 98.1\ N$

The sum of forces needs to equal zero, therefore due to their being two tension wheels, the normal force acting on the drive wheel (T_{n_drive}) is twice the magnitude of the force acting on the tension wheel.

The normal force acting on the guide wheel is found by the calculating the angle from vertical distance (L_V) between the tension and drive wheel and the horizontal displacement (L_H) of the mooring line.

$$\theta = Atan\left(\frac{L_V}{L_H}\right)$$

$$T_n = T_m * \cos(\theta)$$

$$T_{n,drive} = 2 * T_n$$

The table below shows the results from physical tests, where the average minimum force required on the drive wheel is 6 Newtons.

Force Required on Guide Wheel	Force Required on Drive Wheel [N]	Force Required on Drive Wheel + FOS (4) [N]
-3.268184577	-6.536369154	-26.14547662
3.67616562	7.35233124	29.40932496
2.941676385	5.883352769	23.53341108
2.451733889	4.903467778	19.61387111

The test setup for determining the required forces is shown in Figure 33 where the guide wheels (grey) are moved along the slots to find the position where the drive wheel (orange) begins to slip. This distance values are then used to determine the required normal force on the drive wheel and a factor of safety added onto this value.

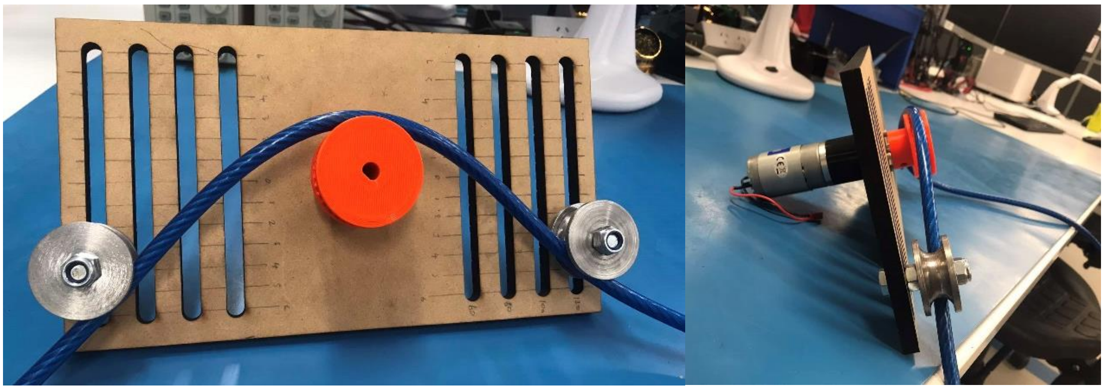


Figure 33: Tension test setup which enabled the optimum tension force to be found.

The 3D printed model of the latest iteration of the lever arm style crawling mechanism is shown in Figure 34 with Figure 35 showing a FEA analysis of the stresses of the lever.

Figure 34: A 3D-printed tensioner arm prototype is undergoing another iteration of development.

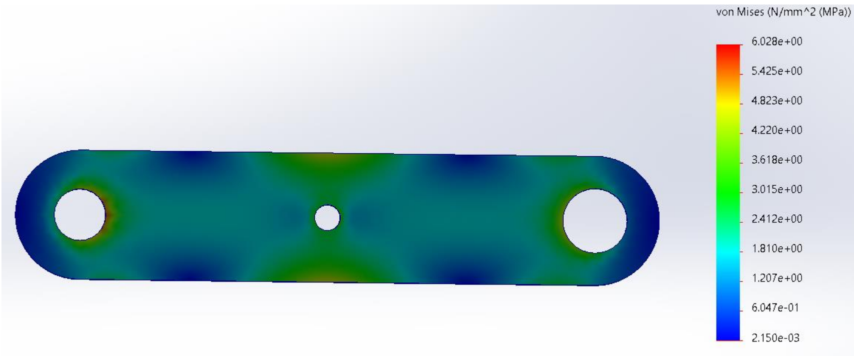


Figure 35: FEA analysis of lever arm shows the arm is sufficient to work with the anticipated loading.

An alternative to the chosen design is a linear compression spring tensioner shown in Figure 36. This design was not the chosen design because of concerns related to its size and complications when in operation.

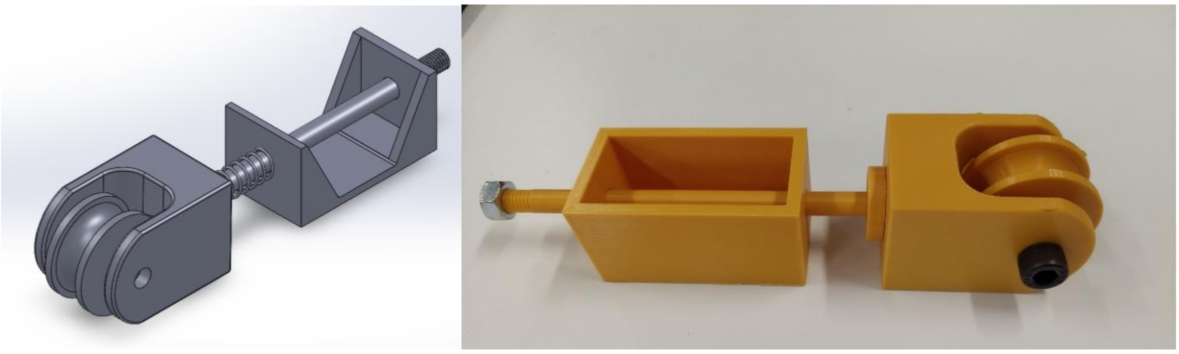


Figure 36: Rejected tensioner concepts which were too bulky and failure-prone for this application.

Another concept used a leaf spring where the guide wheels were attached to the end of the spring, with the centre of the spring pivoting at the drive wheel. To determine whether this option would work, beam bending calculations were used to determine the required moment of inertia of the beam/spring, which could be used to find the width and thickness of the beam (rectangle). But results showed that this was not a viable option due to size constraints of the system, as the spring would be very large to not break. Another limiting factor is the available materials that have a suitable elastic modulus and yield strength relationship that can withstand saltwater conditions. Figure 37 shows the Excel document used to determine the dimensions of the spring.

Constants:			t [m]	W (Corresponding width value) [m]	Max Stress	FOS
Youngs Modulus (E) [N/m^2]	1.965E+11		0.001	0.091345057	147791249	3.38315
Yield Strength [Mpa]	500000000		0.002	0.011418132	295582498	1.69158
			0.003	0.00338315	443373747	1.12772
Changeable Determinants:			0.004	0.001427267	591164996	0.84579
Drive Wheel Force Requirement [N]	30		0.005	0.00073076	738956245	0.67663
Tension Force mooring line Weight [N]	200		0.006	0.000422894	886747494	0.56386
Length Between Drive and Tension Wheels (L) [m]	0.15		0.007	0.000266312	1034538743	0.48331
			0.008	0.000178408	1182329993	0.42289
Essential Calculations:			0.009	0.000125302	1330121242	0.37591
Horizontal Displacement of spring [m]	0.01128177		0.01	9.13451E-05	1477912491	0.33832
Max Force acting on Tension Wheel (F) [N]	15		0.011	6.86289E-05	1625703740	0.30756
Require Moment of Inertia	7.6121E-12		0.012	5.28617E-05	1773494989	0.28193
Determines the Moment of Inertia required for a required force			0.013	4.15772E-05	1921286238	0.26024
on the drive wheel.			0.014	3.3289E-05	2069077487	0.24165
to create a displacement of 50mm. Using a range of thickness			0.015	2.70652E-05	2216868736	0.22554
values the corresponding spring width is determined.			0.016	2.2301E-05	2364659985	0.21145
The maximum stress in the spring is then determined and			0.017	1.85925E-05	2512451234	0.19901
compared			0.018	1.56627E-05	2660242483	0.18795
to the yield strenth to determine the FOS			0.019	1.33175E-05	2808033732	0.17806
			0.02	1.14181E-05	2955824981	0.16916

Figure 37: Spring dimension calculations for sizing the torsion spring for the tensioners.

15.4. Appendix D – Shaft Deflection and O-ring Calculations

Shaft calcs

$D := 8\ \text{mm}$

$A := \pi \cdot \left(\frac{D}{2}\right)^2 = 5.0265 \cdot 10^{-5}\ \text{m}^2$

$F_{wheel} := 30\ \text{N} \quad FOS := 2$

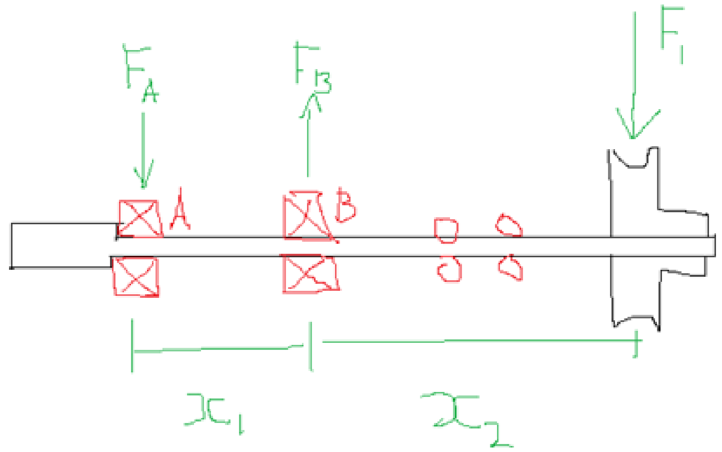
$F_1 := F_{wheel} \cdot FOS = 60\ \text{N}$

$I := \frac{\pi \cdot D^4}{64}$

$E := 193\ \text{GPa}$

$x_2 := 27.5\ \text{mm}$

$x_1 := 20\ \text{mm}$



Seals located at 0.33*x2

Moment at B = 0

$$F_a := \frac{x_2 \cdot F_1}{x_1} = 82.5\ \text{N}$$

Sum of forces = 0

$$F_b := F_a + F_1 = 142.5\ \text{N}$$

Deflections for seal effectiveness

Deflection at wheel

$$\delta_1 := \frac{F_1 \cdot x_2^3}{3 \cdot E \cdot I} = 0.0107\ \text{mm}$$

Deflection at seals (CRITICAL)

$$\delta_1 := \frac{F_1 \cdot \left(\frac{x_2}{3}\right)^3}{3 \cdot E \cdot I} = 0.0004\ \text{mm}$$

Stress in shaft

Max shear stress is at bearing B

$$\sigma_{shear} := \frac{F_b}{A} = 2.8349\ \text{MPa}$$

Bending

$$M_{max} := F_1 \cdot x_2 = 1.65\ \text{J}$$

$$\sigma_{bending} := \frac{M_{max} \cdot \frac{D}{2}}{I} = 32.8257\ \text{MPa}$$

Figure 38: Shaft deflection calculations show that the seals are expected to hold on the motor housing.

From pg 124 (5-18)

Max 92rpm = 9.5 fpm

$FPM = RPM \times \underset{(feet)}{Diameter} \times \pi$

Surface speed = 9.5 fpm

∴ W not critical

Use 1/16 width 0.364"

Add 0.002" to O-ring ID to determine max actual shaft OD

$B = 0.364" + 0.002" = 0.366"^{+0.00}_{-0.001}$ shaft (≈10mm)

Determine gland depth L from 5-4

-012 L_p = 0.065" from pg 145 (5-39)

Gland groove ID A-1 $L_{min} = 0.065$, $L_{max} = 0.067$

$A\text{-}1_{min} = B_{max} + 2L_{min} = 0.366 + 2 \times 0.065 = 0.496$

$A\text{-}1_{max} = B_{min} + 2L_{max} = 0.365 + 2 \times 0.067 = 0.499$

~~= 0.366" + 2 × 0.065" = 0.496"~~

E = 0.012 to 0.016

Shaft bore D

G = 0.075 to 0.079

$D_{min} = 0.366" + 0.012 = 0.378"$

$D_{max} = 0.365 + 0.016 = 0.381"$

Check figure 3-3 (pg 49 (3-3))

Figure 39: Calculations of X-ring glove dimensions and tolerances

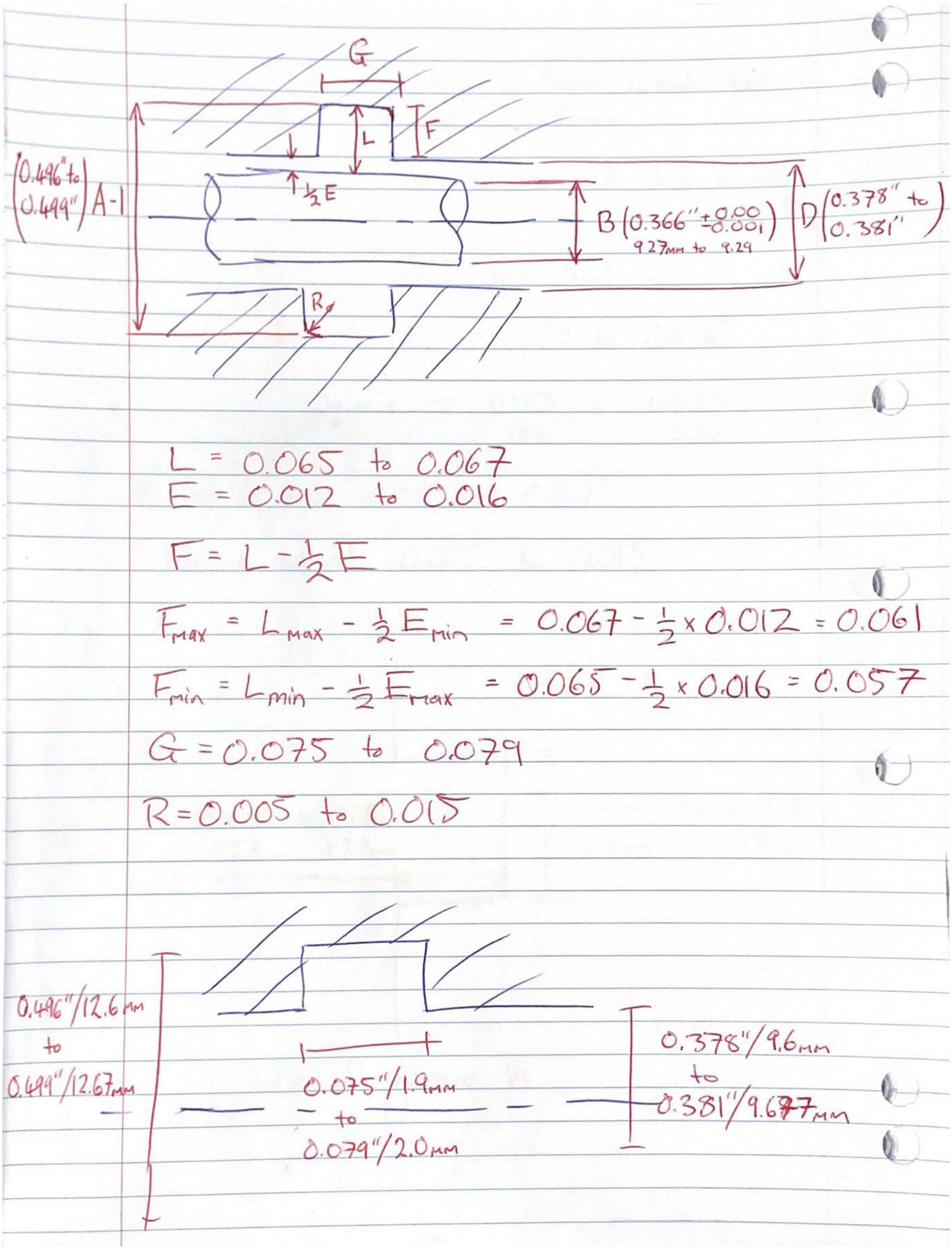


Figure 40: X-ring seal final dimensions and tolerances

15.5. Appendix E - Motor Housing Design Process

Figures 40-42 show motor iterations and Figure 44 shows comparative size reductions between the first and last iterations.

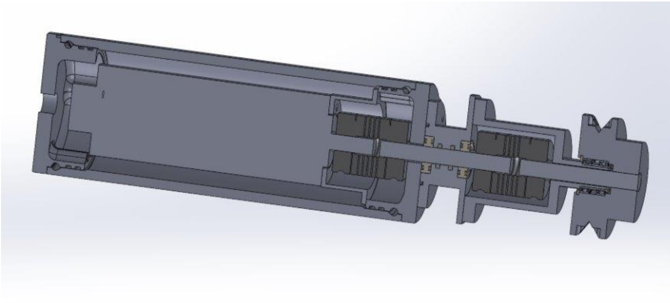


Figure 41: 1st iteration- with axial motor. This was too likely to get caught on obstacles in the ocean. .

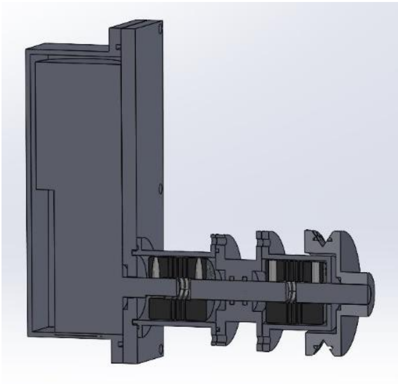


Figure 42: 2nd iteration- with right-angle drive motor. This iteration had unnecessary components.

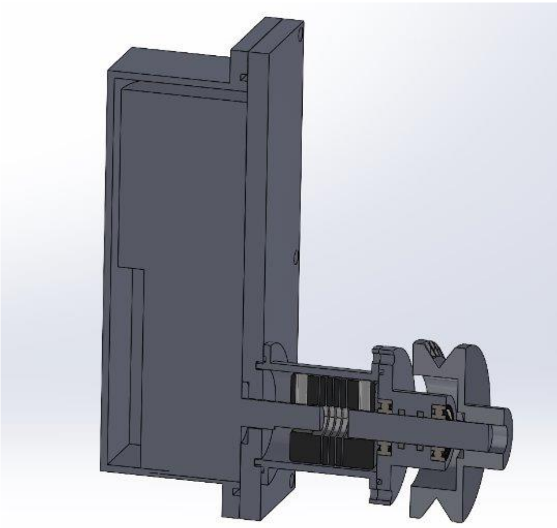


Figure 43: 3rd iteration- This removed an unnecessary bearing and seal. This design was also investigating using a hollowed-out wheel to further reduce the profile which may not be necessary.

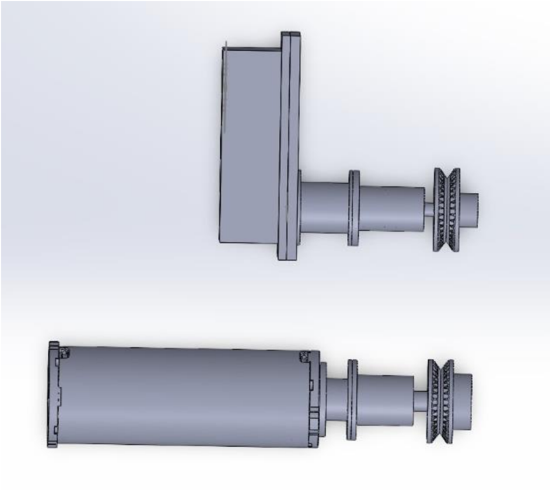


Figure 44: Comparative image showing reduction in drive system horizontal profile:

Figures 44 and 45 show prototypes based on the designs.

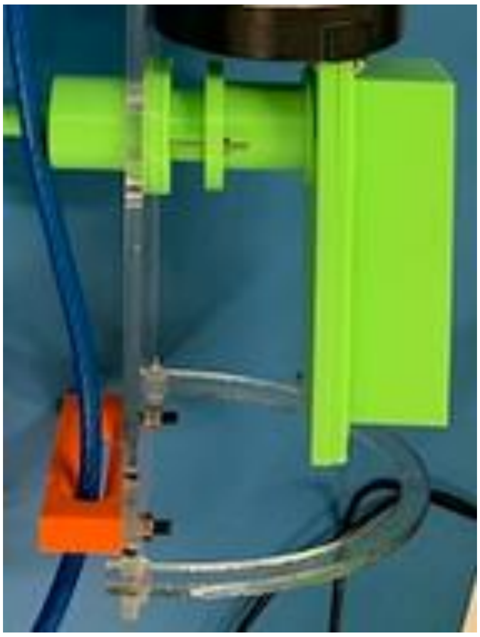


Figure 45: Prototype 1 of the motor housing, based off design iteration #2.

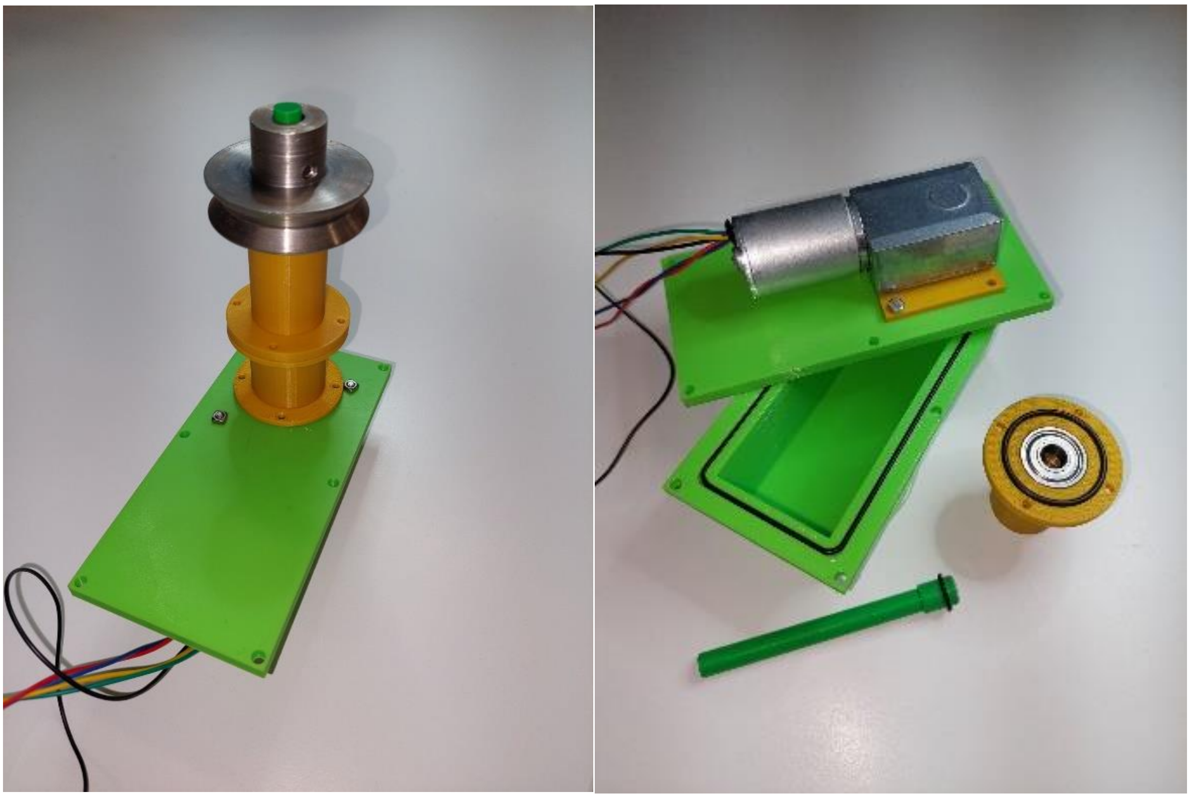
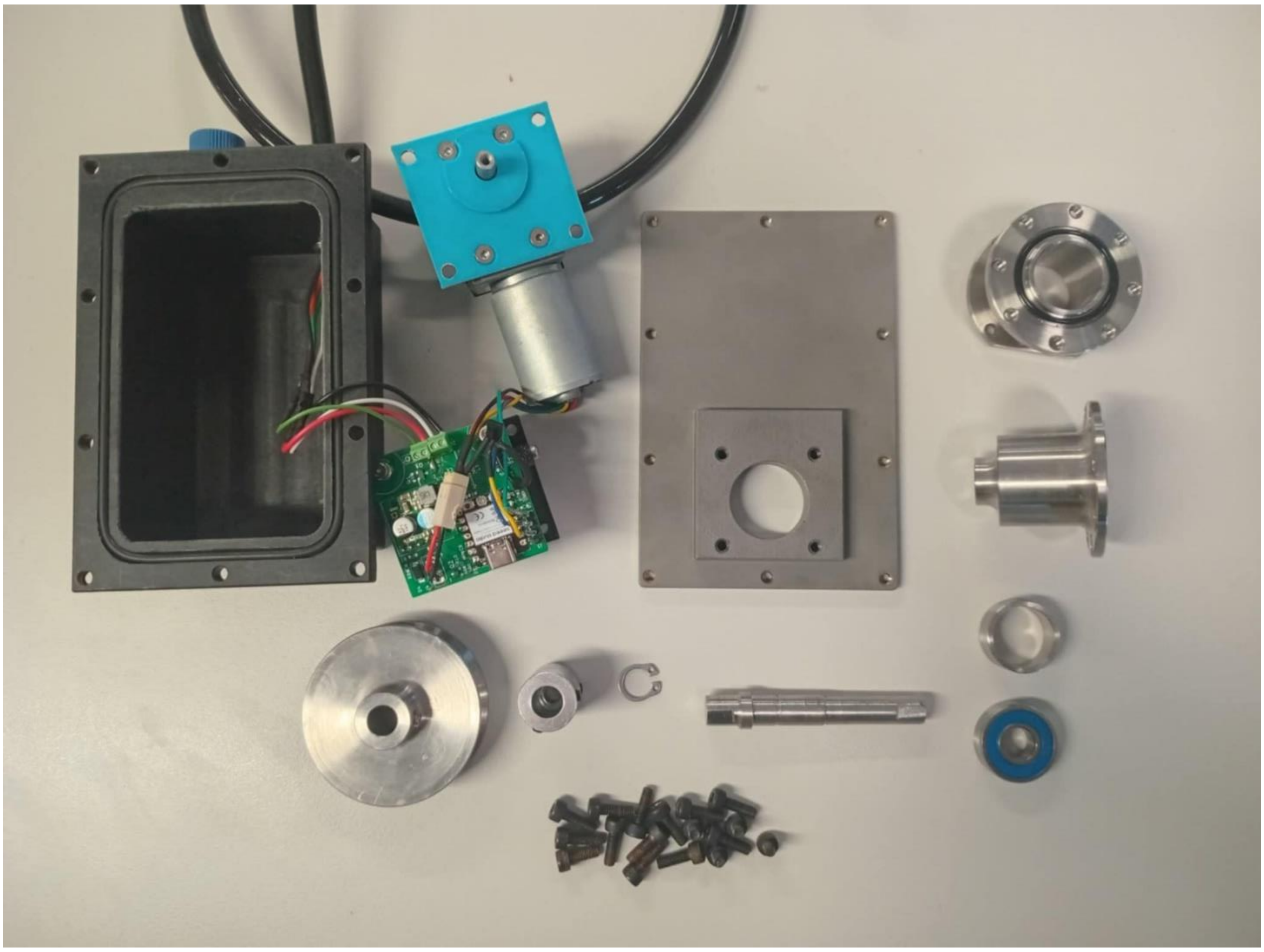


Figure 46: Prototype 2, based off design iteration #3.

15.6. Appendix F – Final Motor Housing Assembly Instructions

M16 Motor Housing Assembly Instructions

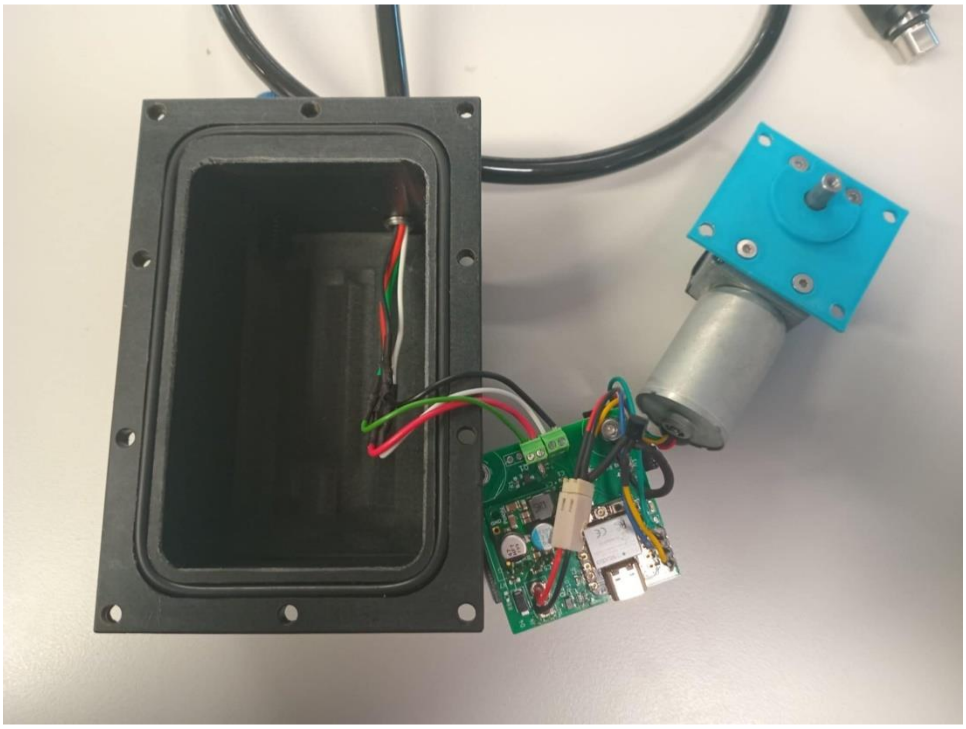
Components required:



Tools required:

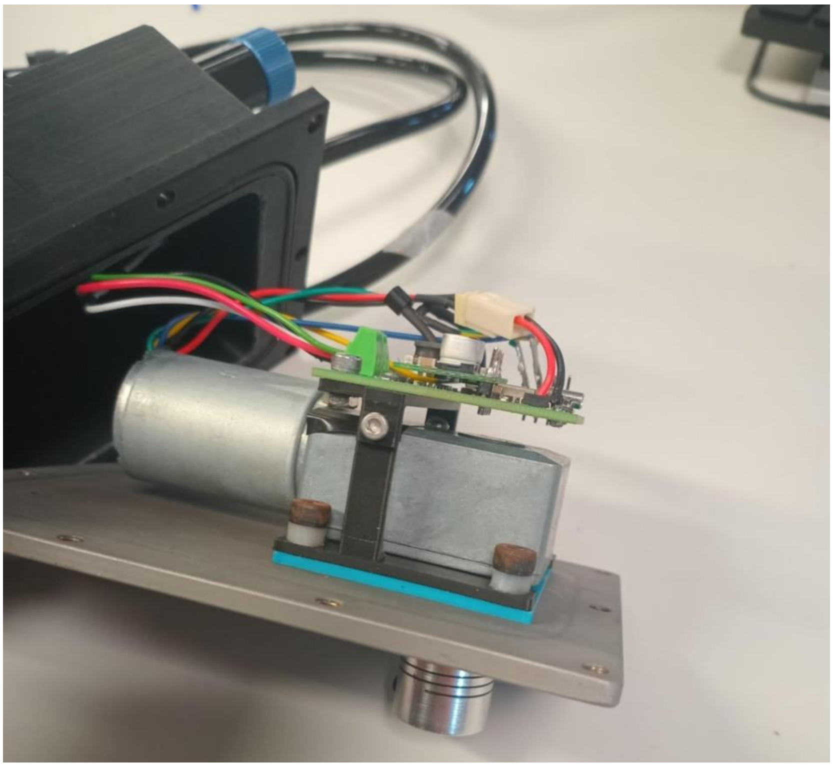
- Circlip pliers for 10 mm Circlip
- 3.0 mm, shortened Allen key
- 2.5 mm Allen key
- 2.0 mm Allen key

Step 1: Wire electrical components into the main motor housing.



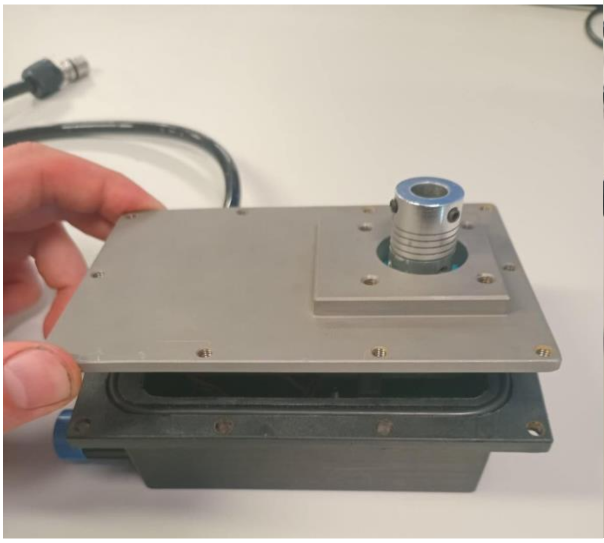
Step 2: Attach flexible coupler to motor shaft with approximately 2mm separation between couple and motor mount (2 mm Allen Key)

Step 3: Mount motor to main Stainless Steel face plate with 4x 12mm M4 Cap screws
(3 mm Allen Key)

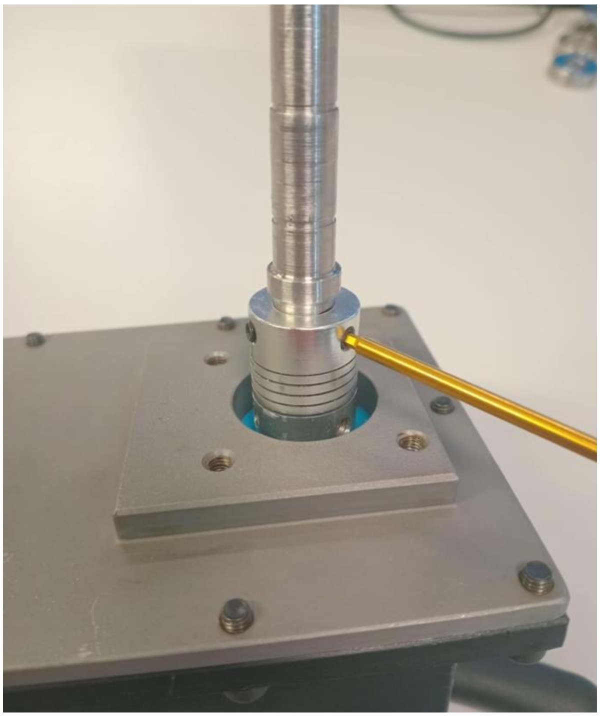


Step 4: Ensure O-ring is installed and mount face plate to acetyl housing with 10 12 mm M4 Cap screws.
(3 mm Allen key)

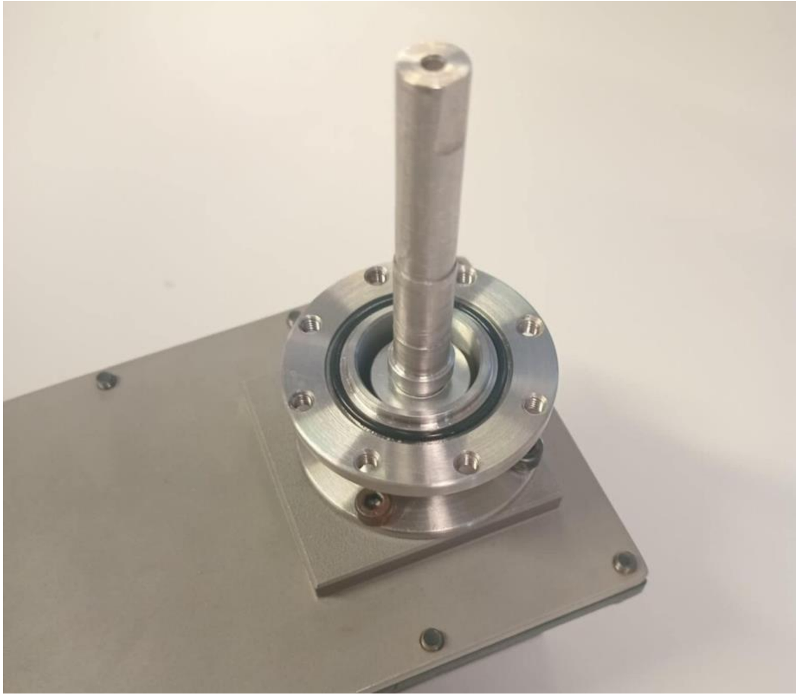
Note: Do not over-tighten as Acetyl may deform



Step 5: Attach drive shaft to coupler, ensuring that when the coupler cover is placed over top, the shoulder on the shaft aligns with the protruding surface of the coupler housing. The flat on the shaft should also align with one of the grub screws.

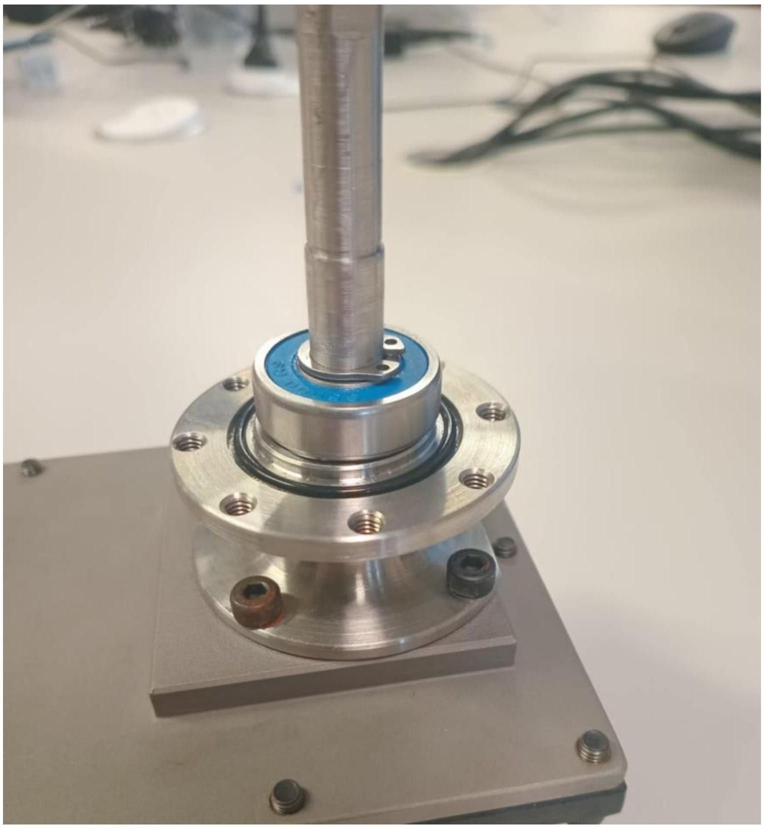


Step 6: Secure coupler cover with 4x 6mm M4 Cap Screws. And O-rig should be installed on each face of the coupler cover. (Modified 3mm Allen Key)

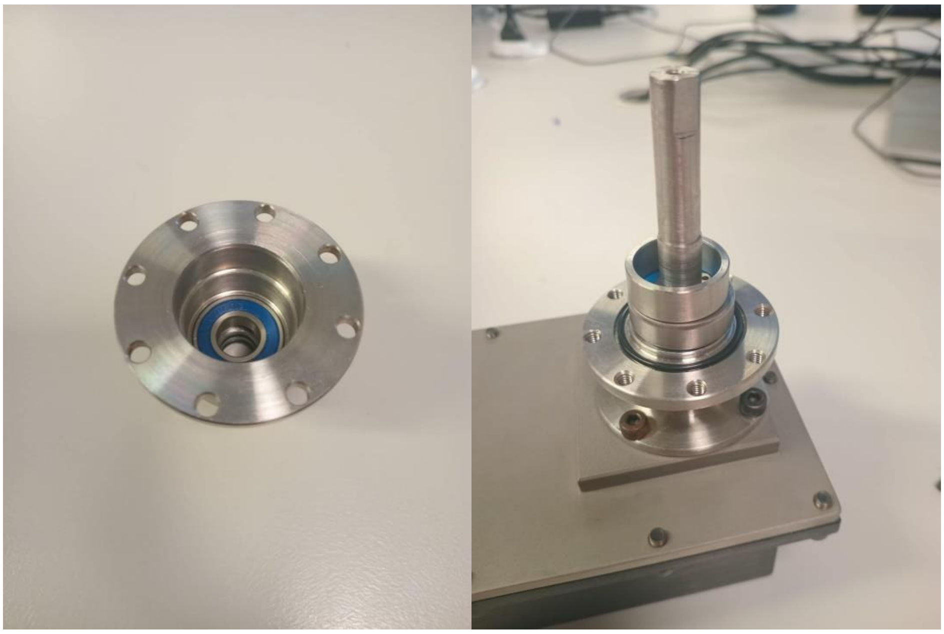


Step 7: Firmly press 26 mm OD bearing onto the shaft by hand until it butts up to the shoulder. The 10 mm circlip can then be installed with the circlip pliers.

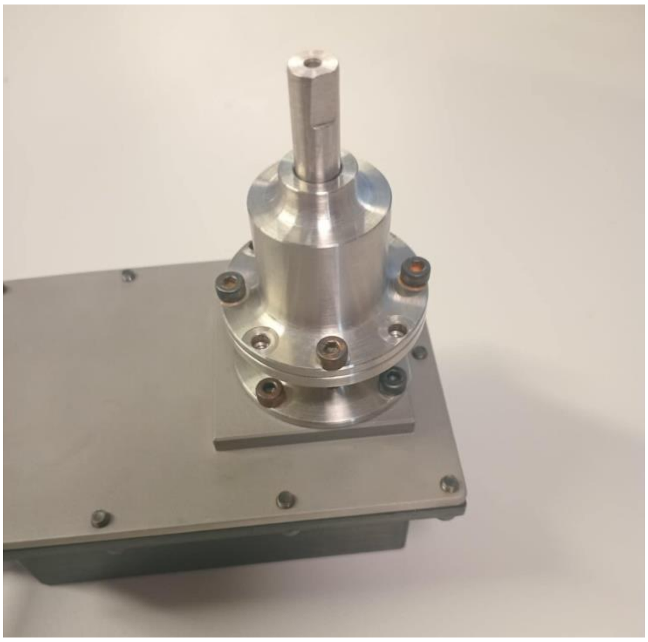
Note: Ensure not to excessively stretch the circlip during installation



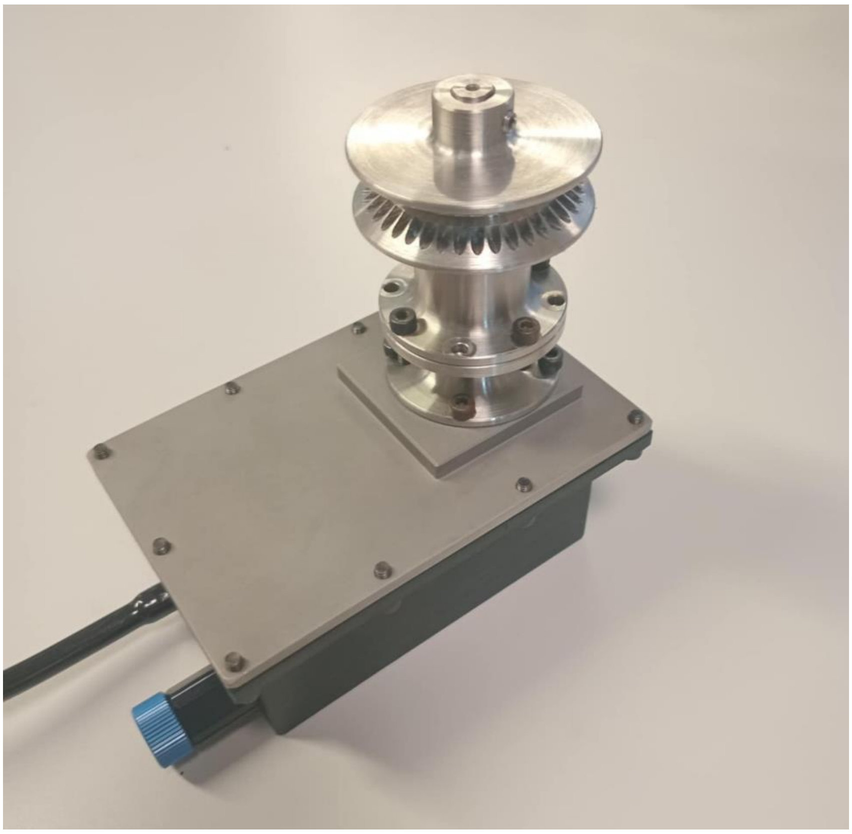
Step 8: Press the 22 mm OD bearing into the bearing housing. Place the spacer on top of the 26 mm OD bearing. 2 X-rings should also be installed in the smaller diameter opening of the bearing housing.



Step 9: The Bearing housing can then be firmly pushed onto the shaft by hand and secured with 4, 12 mm, M4 cap Screws.



Step 10: Place drive wheel on shaft with the flat on the shaft aligned with the grub screw. The position of this can be adjusted based upon the position of the guide wheels.



15.7. Appendix G – Inductive Charging Test Procedure and Results

The aim of these tests was to characterise how the inductive charger performed with different electronic loads and distances between the coils. The inductive receiver was connected to an adjustable electronic load, and the transmitter to an adjustable power supply as shown in Figure 47. The test rig uses a lead screw to vary the distance between coils.

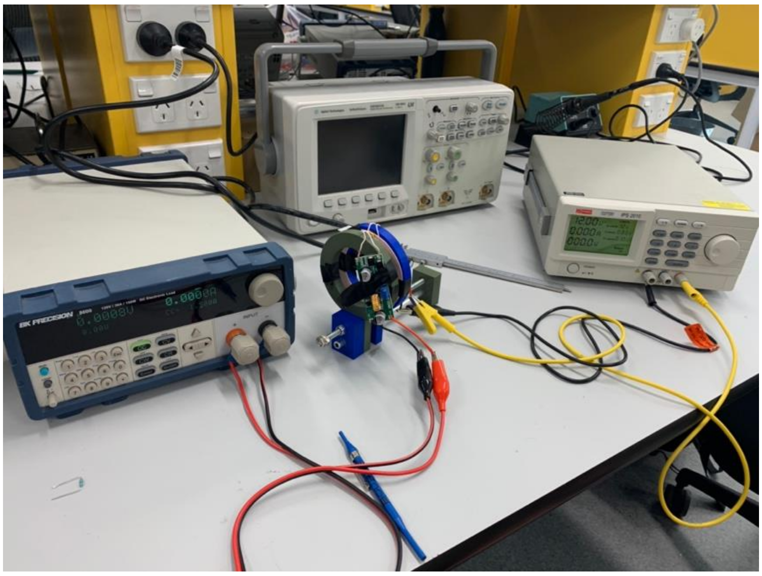


Figure 47: Inductive charger undergoing testing.

Testing was conducted at 12, 18 and 24V input power. The results are plotted below in Figure 48 with efficiency on the top row and power received on the bottom. Efficiency is calculated as input power divided by output power. The red horizontal line is the power necessary to recharge within the given time frame (approximately 8 minutes recharge time to sample every 30 minutes.) The results were fitted with a 3rd order polynomial best fit line.

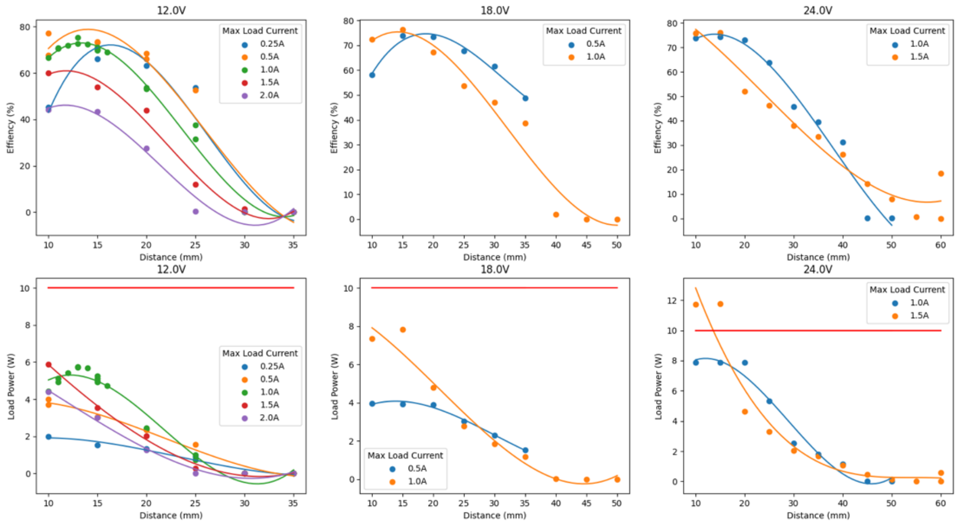


Figure 48: Inductive charging test results. To transfer more than 10 W of power, 24 V was required.

From the results, a 24V input power supply with between 10 to 15mm of distance between coils will provide suitable power transfer. Also, during testing is was noted that the coils lose a lot of efficiency when they are not perfectly parallel. Once the inductive charging pressure housing was designed, the prototype shown in Figure 49 was made from PLA to confirm the coils would transfer enough power at the designed distance.

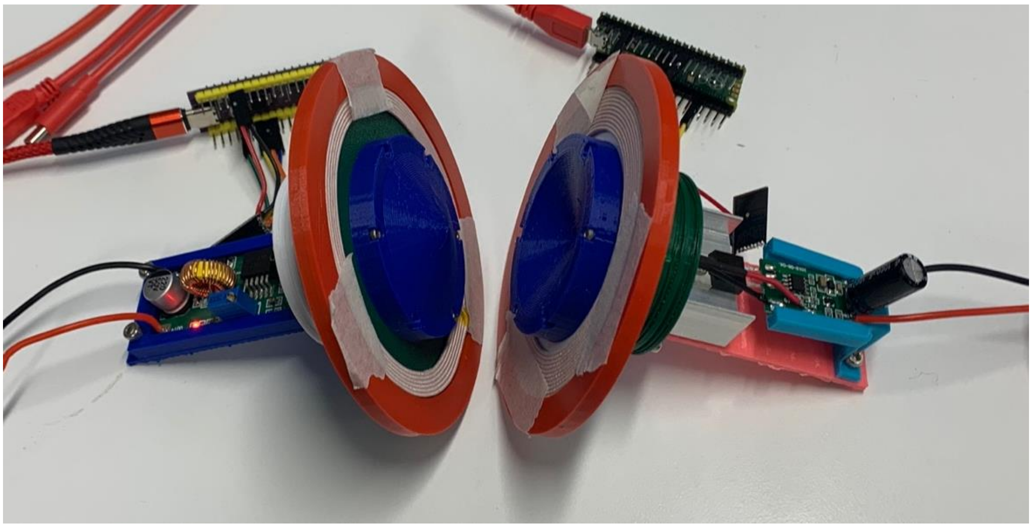


Figure 49: Inductive charger housing prototype, which aims to keep the coils parallel to maximise the efficiencyl.

15.8. Appendix H – Detailed Inductive Charger and Communications Design

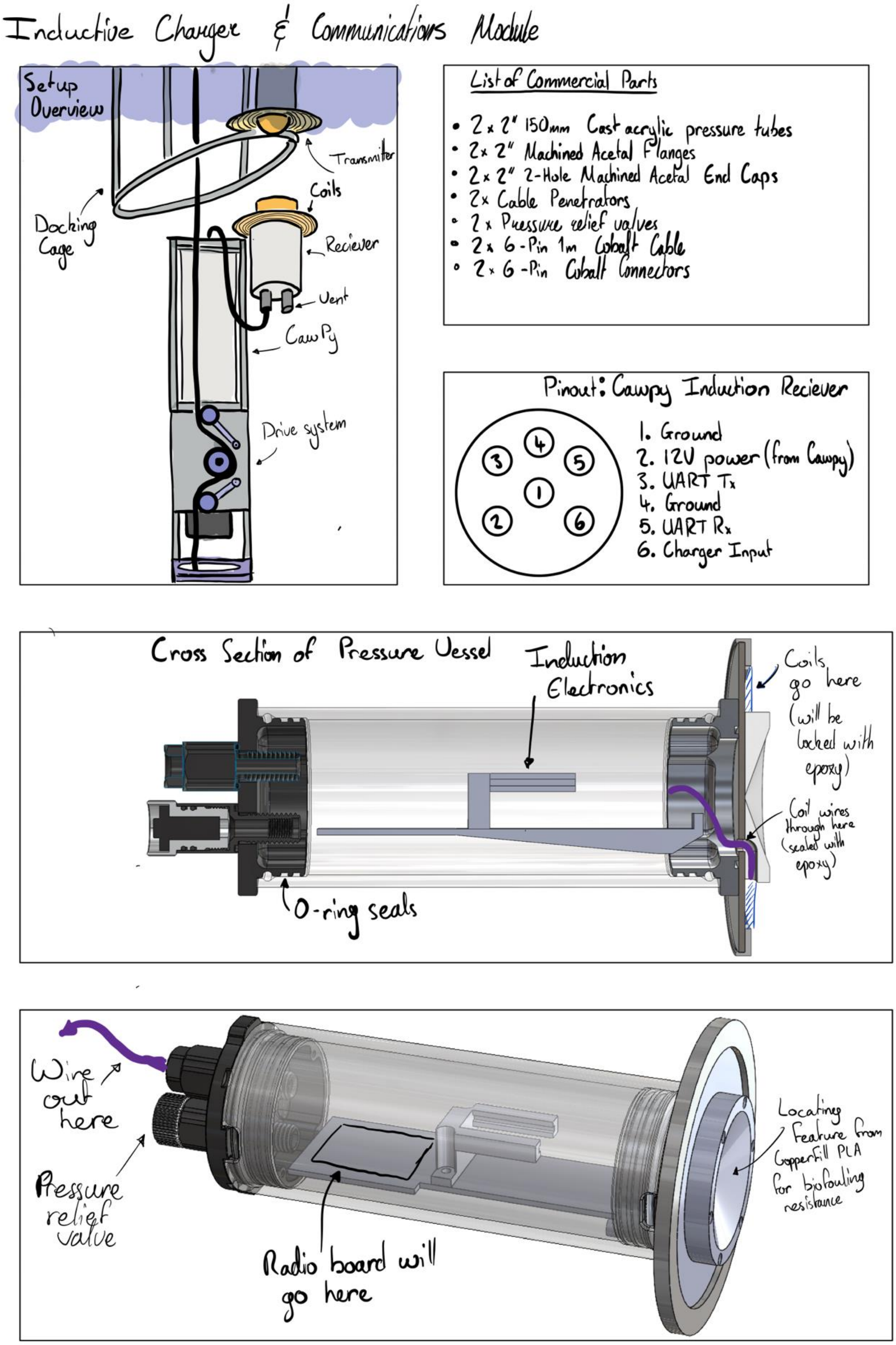


Figure 50: Housing design for inductive charger with radio.

15.9. Appendix I – Custom PCB Schematic

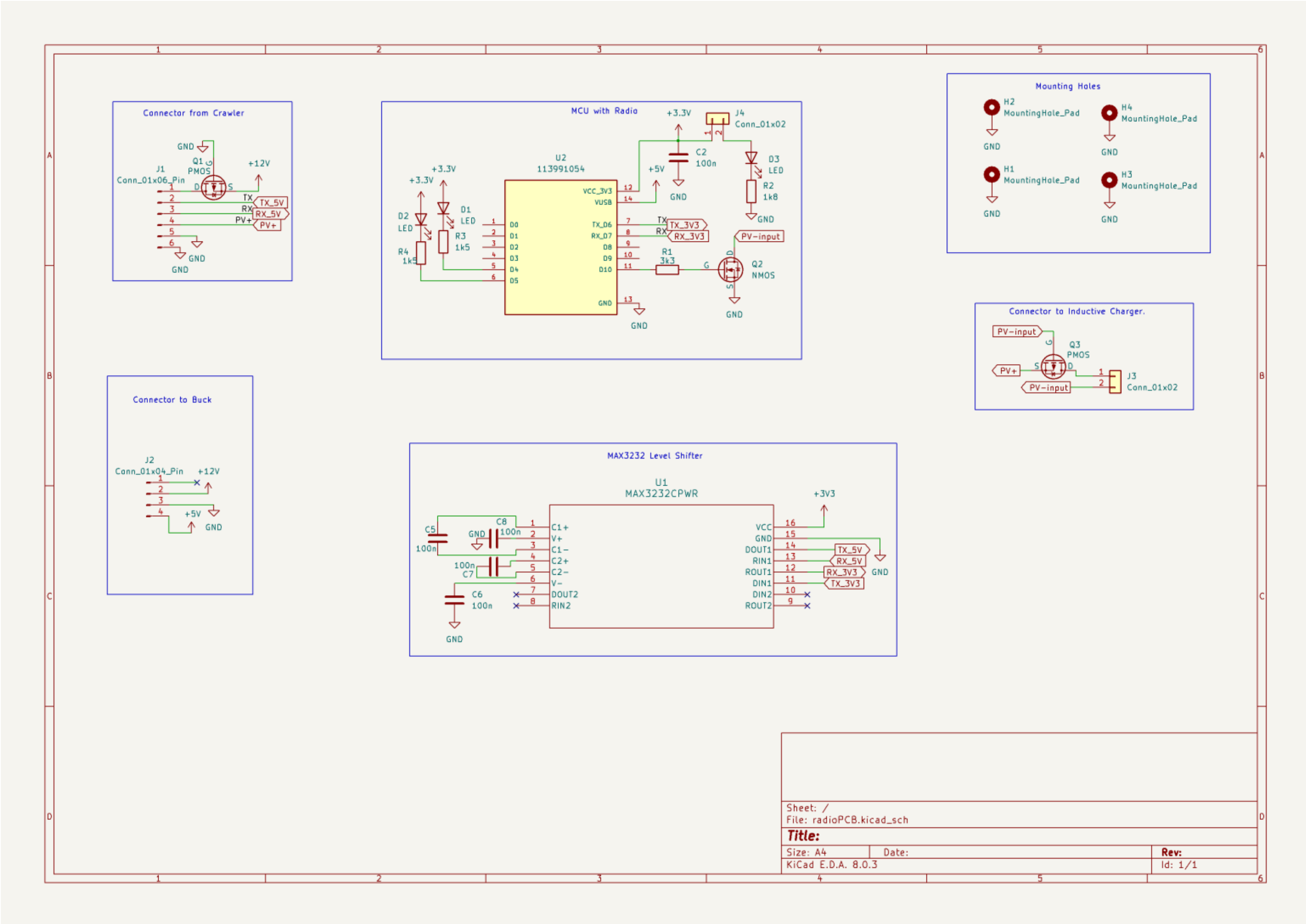


Figure 51: Radio PCB.

15.10. Appendix J: Bluetooth Standards Used

Bluetooth operates on a system of services, characteristics and descriptors. Services identify what the device does, i.e. 'battery monitor' or 'wireless speaker'. The characteristics are individual parameters a paired device can interact with, such as 'battery level' or 'volume level'. Finally, descriptors contain meta-data about services and characteristics, such as units, names, accuracy or timestamps. Each service, characteristic and descriptor has a 16-bit universally unique identification number (UUID). These are detailed in the "Assigned Numbers" Specification here:

https://www.bluetooth.com/specifications/assigned-numbers/

Details on how to programmatically define each characteristic and its' descriptors are found with the Generic Attribute (GATT) supplement:

https://www.bluetooth.com/specifications/gss/

The robot has four services:

- Environmental monitoring service:
  https://www.bluetooth.com/specifications/specs/environmental-sensing-service-1-0-1/
- Automation IO service:
  https://www.bluetooth.com/specifications/specs/automation-io-service-1-0/
- Battery service:
  https://www.bluetooth.com/specifications/specs/battery-service/
- Generic GATT (for robot meta-data), defined in the GATT supplement.

Reading the Bluetooth core specifications may also be useful, but it goes into significant detail unnecessary for implementation considering this project uses the Aioble library.

Aioble Library: micropython-lib/micropython/bluetooth/aioble/README.md at master · micropython/micropython-lib · GitHub

Bluetooth Core Spec: https://www.bluetooth.com/specifications/specs/core-specification-6-0/

Note: these specifications are regularly updated, so care should be taken to ensure the latest version is used.

15.11. Appendix K – Motor Requirement Calculations

Figure 28 depicts motor calculations to verify motor requirements, and Figures 29 and 30 depict calculations to determine those requirements.

Figure 52: Mechanical power and torque curves for the system.

Assume neutrally buoyant

- ~~[illegible]~~ $P = FV$

$$= \frac{1}{2} \rho A v^2 C_d \times v$$

$$= \frac{1}{2} \times 1000 \times 0.5^3 \times C_D \times \frac{\pi \times 0.15^2}{4}$$

$\boxed{P_{v=0.1} = 0.045W.}$ = 12 ~~m~~/5 ~~[illegible]~~ W. ~~[illegible]~~ to overcome drag forces.

This is super tiny, ~~and it will be ha~~ and motors which could only provide this would likely stall from ~~torque~~ starting torque. ~~A~~

12W motor would be sufficient to max this @ ~30-40% efficiency. We'd get 3 V of output power

This gives a maximum movement speed of:

$$\sqrt[3]{\frac{3 \times 2 \times 4}{1000 \times 1.1 \times \pi \times 0.15^2}} = v^m$$

$$v = 0.675\,ms^{-1}$$

max speed.

~~Assume motor~~

If travelling at $0.5\,ms^{-1}$, 60 m journey takes 120 seconds, @0.5A draw.

Requires 120 Aseconds, or 33mAh per trip.

Figure 53: Hand calculations for motor power and power budget.

4 trips per hour gives 240 ~~4?80~~ As, or
66.133 mAh of ~~energy~~ charge used in
that time. This is equivalent to 0.6 wh
of energy per hour. ~~ar~~ Or 0.2 wh
per trip.

~~If char~~
If charging at a rate of 5W @ 80%
efficiency, this will take 5 minutes to
replenish, so in a given hour the ~~pi~~ ~~boat~~
buoy will be charging for ~~B~~ ~~15~~ 40 minutes
and discharging for ~~20~~ 20.

Also, ~~for~~ at a minimum sample rate of
1 trip per hour, to do 72 trips
~~in~~ the onboard batteries need to be
capable of ~~doing~~ storing ~~28.8wh~~ 14.4 wh of energy,

we need ~~a~~ ~~16~~ ~~63~~ 18650 batteries, which will
cost ~~$~~ ~~49.00~~ $57.20

~~5s batteries~~ Use 3 ~~6~~ batteries in a 3S, ~~Pp~~
configuration to get safe current draw
at required capacity.

With 4S, 3P we can use a buck converter
for the motor batteries, which does appeal more
than a boost for some reason.
We will have 15 V to play with and get
40 wh, of capacity, which is 100 hours of operation

Figure 54: Power balance calculations.

15.12. Appendix L – Motor Selection Process

There are a range of motors and motor styles available: brushed DC motors, brushless DC motors, AC motors, stepper motors, and servo motors. AC motors were immediately rejected because it is difficult to power them from DC batteries. Servos and stepper motors were also rejected because of their high current draw which would exceed the battery limits. Therefore, brushed and brushless DC motors were the preferred options.

DC motors are simple, cheap, and effective motors. They have two connections for power and ground, which control the direction the motor spins. The speed of the motor can be controlled by delivering less power to the motor, which is typically done with PWM signals. The motor commutates with brushes which make contact with the rotor. This is a simple system but is susceptible to arcing which can damage the brushes and reduce motor efficiency. These brushes can also wear, reducing the lifespan of the motor. Due to the electrical signals the CawPy can send, external electronics will be required to control the motor.

Brushless DC motors typically have higher efficiency than brushed motors but are more electrically complex. They have supervisory electronics installed to control commutation and provide position feedback. They require four signal lines (power, ground, speed, and direction) and will need simple external electronics to interface with the CawPy. As they are brushless, they have longer lifespans than brushed motors and better performance.

Calculations were undertaken to develop requirements for motor speed and torque of the motors and can be found in Appendix F. The maximum current draw from the motor was developed using power budget calculations from Requirement 19 in Table 1. Other requirements were informed from the project requirements in Table 1. The motor has six requirements which it must satisfy.

1. The motor must be able to move the profiler at a minimum of 40 mm/s.
2. The motor must be able to supply a peak torque of at least 0.1 Nm to overcome any obstacles or currents which may impede the profiler.
3. The average current draw must be fewer than 250 mA to ensure the profiler batteries can last for three days without recharging.
4. The motor must have a lifespan of at least 1500 hours, to ensure that the profiler could operate for up to 6 months at its minimum sampling rates and speeds.
5. The motor must have a right-angled drive for integration into the profiler.
6. The motor must be operated from a DC power supply.

Results from motor testing show that the peak current draw is 2A at 12V and meets the torque requirements. The rated current is 180 mA, and the motor spins at a maximum of 96 RPM, which is sufficient for a travel speed of 40mm/s. The motor can be operated from a DC power supply and can be easily controlled by the CawPy. This motor also has a compact right-angled gearbox installed, which reduces the size and weight of the motor housing.

15.13. Appendix M – Motor Controller Interface Design

The CawPy has 4 bulkheads available for connecting to the motor, these bulkheads output 12 V power, ground, and 5V digital transmit and receive lines that use the UART protocol. The motor needs a PWM signal for speed control and a 3.3V digital signal for direction control. Additionally, the motor should have some hardware such as switches and dials for adjusting speed and direction during testing without needing a computer.

The schematic for the controller prototype is shown in Figure 55 alongside the completed prototype which was constructed from a Raspberry Pi Pico and a breadboard. The voltage divider on the transmit (TX) line converts the incoming commands from 5V logic to 3.3V logic, because the Pico cannot recognise 5V logic. The Pico runs a simple round-robin based task scheduler to read each user input (switches, potentiometer and UART) before sending the updated speed and direction commands to the motor. This program runs automatically when the Pico receives power from USB, meaning no computer is needed to control the motor.

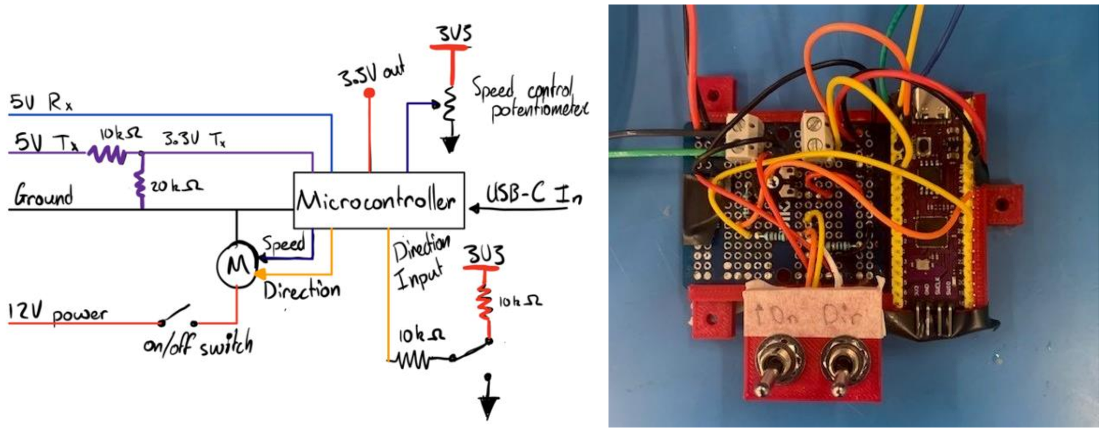


Figure 55: Motor controller schematic (left) and completed prototype (right).

15.14. Appendix N – Recharge Power Calculations

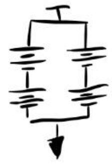
Figure 56 depicts the charging requirements for the inductive charger. With 10W of charging, 1/6th of the batteries can be recharged each run, which is recharging more than the motors use.

Recharge Power Calculations

Battery Stats:

Model: IFR26650-38A
Capacity: 3800 mAH (0.5 C)
Voltage: 3.2V
Charge Current: 1900mA (standard) 3800mA (max)

Configuration: 2 series 2 parallel →



⇒ Effective voltage = 6.4 V,
⇒ Effective capacity = 7600 mAH

A 0.5 C rating means batteries take (2 hours) 120 minutes to fully charge.
We have 40 minutes to recharge so at most (40/120 = 1/3) of the battery capacity (2534 mAH) can be recharged.

Achieving this in 40 minutes requires

$$\frac{2534\,mAH}{\frac{2}{3}H} = 3801\ mA \text{ or } 3.8A$$

The batteries need 3.65 volts to charge, so

3.65 × 3.8 = 13.87 Watts of power at the inductive reciever.

"However 10W of power should be sufficient" (Cawthron)

Assuming the charger is 80% efficient,

$$\frac{13.87}{0.8} = 17.34 \text{ Watts of power at the inductive transmitter}$$

Figure 56: Inductive power transfer calculations to determine the minimum charging rate.